

GOVERNMENT OF JHARKHAND WATER RESOURCES DEPARTMENT

REQUEST FOR PROPOSAL (RFP)

**Selection of Agency for Re-Design, Re-Development, Implementation, Operation,
and Hosting of an Integrated Software Solution for the Water Resources
Department, Government of Jharkhand**

comprising

- 1. Project Progress Monitoring System (PPMS)**
- 2. Contractor Registration & Empanelment Portal**
- 3. Industrial Water Allocation Portal**
- 4. Water E-Tariff & Billing Portal**
- 5. CMS-based Departmental Website**

along with Comprehensive Maintenance and Support for a period of three (3) years.

NIT NO.: WRD/JH/CE/DMPH/RFP/____/2025-26

Mode of Bidding: National Competitive Bidding (Online)

Issued by:

Office of the Executive Engineer

Water Ways Division, Ranchi

Water Resources Department, Government of Jharkhand

Nepal House, Doranda, Ranchi - 834002

TABLE OF CONTENTS

DISCLAIMER.....	6
NOTICE INVITING REQUEST FOR PROPOSAL (RFP)	7
Critical Dates and Bid Information.....	7
Key Terms of the Notice Inviting Tender	8
SECTION-I: INSTRUCTIONS TO AGENCY/BIDDER/CONSULTANT	10
1. Minimum Eligibility Criteria	10
1.1 Technical Requisites.....	10
1.1.2 Eligibility Criteria of the Agency/Bidder/Consultant.....	10
1.1.3 Past Project Experience.....	10
1.1.4 Statutory Registrations.....	10
1.1.5 Empanelment with Government Bodies	10
1.1.6 Affidavit of Non-Conviction and Non-Blacklisting.....	10
1.2 Financial Requisites.....	11
1.3 Participation	11
2. Language of the Bid	11
3. Pre-Bid Meeting	11
4. Amendment of Bid.....	12
5. Preparation of the Bid.....	12
6. Department's Right to Accept Any Bid and to Reject Any or All Bids	13
7. Performance Guarantee	13
8. Signing of Agreement.....	13
9. Time Schedule	13
10. Information for Agency/Bidder/Consultant for Submitting the Bid.....	13
11. Technical Bid.....	14
11.2 Mandatory Key Personnel.....	15
12. Opening of Bid	15
13. Evaluation of the Bid	15
13.1 Evaluation of the Technical Bid	16
13.2 Evaluation of the Financial Bid.....	17
14. Negotiations and Weight of Bid	18
15. Substitution of Key Personnel.....	18
16. Payment Schedule	18
17. Compliance with Labour Regulations	19
18. Hand Over and Take Over (HOTO) – Intellectual Property Rights and Source Code	19
18.1 Intellectual Property Rights (IPR).....	19
18.2 Source Code and Technical Artifacts	20
18.3 Documentation Requirements.....	20

18.4 Training Requirements.....	20
18.5 HOTO Process	21
18.6 Post-Handover Support	21
18.7 Exit Management.....	21
19. General Provisions	21
SECTION-II: SPECIAL CONDITIONS	22
1. Public Acceptability of the Project.....	22
2. Delays in Performance	22
3. Liquidated Damages	22
3.4 SLA-Based Penalties During AMC / Operations.....	23
4. No Claim Arrangements	23
5. Quality Standards.....	23
6. Confidentiality and Data Protection	24
7. Transfer of Rights.....	24
8. Decisions and Interpretation	24
9. Suspension	24
10. Termination of the Contract	24
11. Cessation of Rights and Obligations.....	25
12. Cessation of Services Upon Termination.....	25
13. Dispute Resolution	25
14. Force Majeure	25
15. Contract Validity and Extension.....	26
SECTION-III: TERMS OF REFERENCE (TOR) AND SCOPE OF WORK	27
1. Introduction	27
2. Overall Objectives	28
3. Inadequacies in the Current Software / Modifications Required.....	28
4. Overall Scope of Work — Composite Project	29
5. Component-A: Project Progress Monitoring System (PPMS)	29
5.1 Objective.....	30
5.2 Scope of Work — Component-A (PPMS).....	30
5.3 Project Initiation	30
5.4 Requirements Analysis	31
5.5 System Design and Architecture	31
5.6 Development.....	31
5.7 Testing.....	31
5.8 Deployment	31
6. Modules for the PPMS Portal	31
6.1 Module A — Fund Requisition Module	31
6.2 Module B — Reporting Management Module.....	33

6.3 Module C — Customisation of MIS & Dashboard.....	33
6.4 Module D — BI-Enabled Dashboard	33
9. Component-B: Contractor Registration & Empanelment Portal — Redesign & Redevelopment (wrddreg.jharkhand.gov.in).....	34
9.1 Objective.....	34
9.2 Portal Redesign & UI/UX Overhaul	34
9.3 Public Portal Modules	34
9.4 Department Admin Panel Modules.....	35
9.5 Additional Modules	36
8. Component-C: Industrial Water Allocation Portal — Re-engineering & Modernisation (wrdd.jharkhand.gov.in).....	38
8.1 Objective.....	38
8.2 Core Functional Modules	38
8.3 Integration Requirements.....	40
8.4 Report Modules	40
10. Component-D: Water E-Tariff & Billing Portal.....	41
10.1 Project Background and Objective	41
10.2 Scope of Work — Component-D.....	41
10.3 Functional Modules.....	41
10.4 Role-Based Access and Workflow System	42
10.5 Tariff and Billing Module.....	43
10.6 Approval Workflow	44
10.7 Payment Integration Module	44
10.8 Revenue and Recovery Management.....	45
10.9 Dashboard and MIS	45
10.10 Security Requirements (Component-D Specific)	46
10.11 GIGW Compliance	46
10.12 Security Audit.....	46
10.13 Support and Maintenance (Component-D).....	47
10.14 Training (Component-D)	47
10.15 Deliverables (Component-D).....	47
10.16 Ownership.....	47
10.17 Implementation Timeline (Component-D).....	48
10.18 Expected Benefits.....	48
11. Component-E: CMS-based Departmental Website.....	49
11.1 Project Background and Objective	49
11.2 Scope of Work — Component-E	49
11.3 Website Design & UI/UX Requirements	50
11.4 Public-Facing Modules.....	51

11.5 Content Management System (CMS) Requirements.....	53
11.6 Integration Requirements (Component-E).....	54
11.7 Bilingual & Accessibility Requirements.....	54
11.8 Citizen Engagement & Analytics	55
11.9 Content Migration.....	55
11.11 Security Requirements (Component-E).....	55
11.12 Hosting, Backup & Disaster Recovery (Component-E)	56
11.13 Training (Component-E).....	56
11.14 Deliverables (Component-E)	56
11.15 Implementation Timeline (Component-E)	56
11.16 Ownership (Component-E)	57
11.17 Expected Benefits (Component-E).....	57
12. Data Migration	58
12.1 Scope of Data Migration	58
12.2 Migration Methodology.....	58
13. Technology Stack & Technical Requirements	59
14. Security Requirements & Compliance.....	60
14.1 Application Security	60
14.2 Infrastructure Security	61
14.3 Regulatory & Policy Compliance	61
15. Hosting, Deployment & Infrastructure.....	61
16. Timelines & Deliverables.....	62
16.1 Macro Timeline	62
16.2 Master Deliverables	63
17. Project Management Unit (PMU), Operation & Maintenance	64
17.1 Activities During AMC	64
17.2 Mandatory On-Site Key Personnel.....	64
17.3 Roles & Responsibilities — Key Roles	64
17.4 SLA Framework During AMC.....	65
Annexure-1: Format for Financial Bid.....	66
Annexure-2: Performance Bank Guarantee	69
Form T-1: Agency/Bidder/Consultant Details	71
Form T-2: Past Experience	73
Form T-3: Personnel Qualification & Competence	74
Form T-4: Format of Curriculum Vitae (CV) for Proposed Key Staff	75
Form T-5: Affidavit	77

DISCLAIMER

1. Though adequate care has been taken while issuing this Bid Document, the Agency/Bidder/Consultant should satisfy itself that the document is complete in all respects. Intimation of any discrepancy shall be given to the issuing office immediately. If no intimation is received within 3 (three) days from the date of issue of the Bid Document, this office shall consider that the document received by the Agency/Bidder/Consultant is complete in all respects.
2. The Water Resources Department, Government of Jharkhand reserves the right to change any or all of the provisions of this Bid Document before the date of submission. Such changes will be intimated on the e-tender website www.jharkhandtenders.gov.in. It will be the responsibility of the Agency/Bidder/Consultant to visit the website regularly to keep itself updated.
3. The Water Resources Department reserves the right to reject any or the entire bid without assigning any reason whatsoever. No correspondence will be entertained on this account.
4. Information furnished in this RFP is intended only as a guide and shall not be binding on the Department in any manner. Bidders are expected to make their own independent assessment before submitting their proposal.

NOTICE INVITING REQUEST FOR PROPOSAL (RFP)

WATER RESOURCES DEPARTMENT, GOVERNMENT OF JHARKHAND

National Competitive Bidding

Critical Dates and Bid Information

S. No.	Particulars	Details
1	Name of the Work	Selection of Agency for Re-Design, Re-Development, Implementation, Operation, and Hosting of an Integrated Software Solution for the Water Resources Department, Government of Jharkhand, comprising (i) Project Progress Monitoring Portal (PPMS), (ii) Contractor Registration & Empanelment Portal, (iii) Industrial Water Allocation Portal, (iv) Water E-Tariff & Billing Portal, and (v) CMS-based Departmental Website, along with Comprehensive Maintenance and Support for a period of three (3) years.
2	NIT Number	WRD/JH/CE/DMPH/RFP/_____/2025-26
3	Mode of Submission of Bids	Online through www.jharkhandtenders.gov.in
4	Tender Fee (Non-refundable)	Rs. 10,000.00 (Rupees Ten Thousand only) — Online Mode through Portal. Jharkhand MSME and Startup DPIIT-recognized entities are exempted from paying the Tender Fee.
5	Earnest Money Deposit (EMD) / Bid Security	Rs. 1,00,000.00 (Rupees One Lakh only) — Online Mode through Portal. Jharkhand MSME and Startup DPIIT-recognized entities are exempted from paying the Tender Fee.
6	Time of Completion	48 Months (4 years) from the date of Letter of Acceptance (LOA) — comprising 12 months for Design, Development, Testing, Hosting and Go-Live, followed by 36 months of Operation, Maintenance, and Annual Maintenance Contract (AMC).
7	Period of Availability of Bidding Document on Website	From: _____ To: _____
8	Last Date and Time for Submission of Pre-Bid Queries	Date: _____ Time: _____ Hrs
9	Date, Time and Place of Pre-Bid Meeting	Date: _____ Time: _____ Hrs Venue: Conference Hall, 1st Floor, Nepal House, Doranda,

S. No.	Particulars	Details
		Ranchi, Jharkhand - 834002
10	Last Date and Time for Online Submission of Bids	Date: _____ Time: _____ Hrs
11	Date, Time and Place of Opening of Technical Bid	Date: _____ Time: _____ Hrs Venue: Office of the Executive Engineer, Water Ways Division, Ranchi
12	Date, Time and Place of Opening of Financial Bid	To be intimated to technically qualified bidders
13	Bid Validity Period	120 (One Hundred Twenty) days from the last date of bid submission
14	EMD Validity Period	45 (Forty-Five) days beyond the validity of the bid
15	Helpline / Helpdesk Number	_____

Note: All communications regarding this tender should be addressed to: Executive Engineer, Water Ways Division, Jal Bhawan, Doranda, Ranchi - 834002, Jharkhand. For further details, please refer to www.jharkhandtenders.gov.in.

Key Terms of the Notice Inviting Tender

1. The Water Resources Department, Government of Jharkhand invites online proposals from eligible Agencies / Bidders / Consultants of national and international repute having adequate experience in the execution of Government software development projects in India for the Selection of Agency for Re-Design, Re-Development, Implementation, Operation, and Hosting of an Integrated Software Solution for the Department.
2. All Agencies/Bidders/Consultants must read and comply with the Minimum Eligibility Criteria, Terms of Reference, Scope of Work and other conditions set out in this RFP.
3. Bids shall be submitted only online through the e-procurement portal www.jharkhandtenders.gov.in. Physical / hard copy submission shall not be accepted.
4. The Tender Fee and EMD/Bid Security shall be payable online through the e-procurement portal in adherence to Memo No. 120 dated 03.10.2023 of the Information Technology & e-Government Department, Government of Jharkhand.
5. Agencies/Bidders/Consultants shall upload the scanned and signed copies of the Bid Document, Annexure-III (Form T-5 - Affidavit), and Annexure-I (Financial Bid Form), failing which the bid may be rejected without further notice.
6. For evaluation of the Technical Bid, one representative each from the National Informatics Centre (NIC), Jharkhand and the Jharkhand Agency for Promotion of Information Technology (JAPIT) shall be members of the Departmental Tender Evaluation Committee.

7. The Financial Bid shall be opened only for those Agencies/Bidders/Consultants who qualify in the Technical Bid.
8. Quality assurance of the work assigned to the successful Agency/Bidder/Consultant shall be ensured at regular intervals by a committee comprising of: (i) two nominated officials from WRD, Government of Jharkhand; (ii) one representative of NIC, Jharkhand; and (iii) one representative of JAPIT, Jharkhand.
9. The Water Resources Department reserves the right to accept or reject any proposal in part or in whole without assigning any reason thereof.
10. The successful Agency/Bidder/Consultant must have a local office located at Ranchi. Agencies/Bidders/Consultants who do not currently have a local office at Ranchi shall establish such office within two (2) months from the date of issue of the Letter of Acceptance (LOA).
11. Bids without the requisite details/documents are liable for rejection, and the Department will not entertain any further communication in such cases.

Executive Engineer

Water Ways Division, Ranchi

Water Resources Department, Government of Jharkhand

SECTION-I: INSTRUCTIONS TO AGENCY/BIDDER/CONSULTANT

1. Minimum Eligibility Criteria

1.1 Technical Requisites

1.1.1 In order to qualify for the Technical Bid, the Agency/Bidder/Consultant must be:

- A Company registered under the Companies Act, 1956 or 2013, OR
- A Partnership firm registered under the Partnership Act, 1932, OR
- A Proprietorship Firm, OR
- A Limited Liability Partnership registered under the Limited Liability Partnership Act, 2008.

Certificate of Incorporation and all relevant documents with statutory operational permission in India are required to be submitted with the bid.

1.1.2 Eligibility Criteria of the Agency/Bidder/Consultant

The Consultant/Bidder/Agency shall fulfil the prescribed technical and financial eligibility criteria specified under this RFP. The Consultant/Bidder/Agency shall be engaged in the business of software development and website/web portal development, and shall possess demonstrable experience in the execution of similar assignments.

1.1.3 Past Project Experience

The Bidder must have completed at least one (1) project of value Rs. 6.00 Crore (Rupees Six Crore) or above for Website Development/Web Application / Software / Web Portal in any Government Sector / Public Sector Undertaking (PSU) in the last three (3) financial years (2022-23, 2023-24, 2024-25). The Bidder must attach a Certificate of Completion issued by the competent authority of the concerned Government Sector / PSU.

1.1.4 Statutory Registrations

- The Bidder shall hold a valid Permanent Account Number (PAN) issued by the Income Tax Department.
- The Bidder shall hold a valid Goods and Services Tax (GST) Registration Number.

1.1.5 Empanelment with Government Bodies

The Bidder shall be empanelled with any State Government Department, Central Government Ministry/Department, Public Sector Undertaking (PSU), or other Government body for the design, development, implementation, operation, and maintenance of application software and websites. Documentary evidence of such empanelment shall be submitted along with the bid.

1.1.6 Affidavit of Non-Conviction and Non-Blacklisting

The Bidder/Agency/Consultant shall submit, along with the Part-I Technical Bid, an Affidavit on a Non-Judicial Stamp Paper of Rs. 100/- duly notarised by a Notary Public, declaring that the Bidder/Agency/Consultant or any of its Directors / Office Bearers / Authorised Agents:

- Has not been convicted by any court of law;

- Has not been blacklisted by any Government / Semi-Government concern;
- Does not have any named FIR or Criminal Case regarding misappropriation / defalcation of Government funds pending against it or its sister concerns by any authority.

The format of the said Affidavit is provided at Form T-5 of this RFP.

1.2 Financial Requisites

1.2.1 The Bidder/Consultant must have achieved a minimum average annual turnover of INR 8 (Eight) Crore in the last three (3) financial years (2022-23, 2023-24, 2024-25). The Bidder must attach a Certificate of Annual Turnover issued by a Chartered Accountant using a valid UDIN.

1.3 Participation

1.3.1 The Bidder/Agency/Consultant may submit its proposal as a single entity. The Bidder shall be fully responsible for all contractual, technical and commercial obligations arising out of this RFP.

1.3.2 The Bidder shall fulfil the prescribed technical and financial eligibility criteria specified under this RFP and shall be actively engaged in the domain of software development, website/web portal development, IT services, with demonstrable experience in the successful execution of similar assignments.

1.3.3 In order to promote local participation and strengthen the IT ecosystem of Jharkhand, preference shall be given to GM(DIC) verified Jharkhand-based MSME units having prior experience in the execution of projects for the Government of Jharkhand. Such bidders shall submit supporting documentary evidence, including a minimum of three (3) or more government similar work orders along with corresponding completion or performance certificates.

1.3.4 All applicable MSME benefits, concessions, and exemptions shall be extended in accordance with the provisions of the Jharkhand Procurement Policy, 2014, and amendments issued thereunder from time to time.

1.3.5 All certificates and documents related to the financial details of the Bidder/Agency/Consultant shall be signed through a Chartered Accountant or Company Secretary holding a valid UDIN.

2. Language of the Bid

Agencies/Bidders/Consultants are required to furnish all information and documents called for in this Document in the English language only.

3. Pre-Bid Meeting

3.1 A Pre-Bid Meeting shall be convened at the designated date, time and place as indicated in the Notice. A maximum of two (2) representatives of each Bidding firm shall be allowed to participate on production of an authority letter from the Bidding firm.

3.2 During the course of the Pre-Bid Meeting, Bidding firms will be free to seek clarifications and make suggestions for consideration of the Procuring Authority. The Authority shall endeavor to provide clarifications and such further information as it may, in its sole discretion, consider appropriate for facilitating a fair, transparent, and competitive Selection Process.

4. Amendment of Bid

At any time prior to the deadline for submission of bids, WRD, Government of Jharkhand may, for any reason, whether on its own initiative or in response to clarifications requested by a Bidding firm, modify the bid document by issuance of an Addendum/Amendment and posting it on the e-tender website (www.jharkhandtenders.gov.in). Bidders are advised to regularly visit the site to keep themselves apprised of such amendments.

5. Preparation of the Bid

5.1 The Bid shall be signed by the Agency/Bidder/Consultant or by a duly authorized person holding power of attorney for signing the bid document. A copy of the power of attorney shall be submitted with the bid.

5.2 In case the bid is submitted by a proprietary firm, it shall be signed by the proprietor giving full name and the seal of the firm.

5.3 In case the bid is submitted by a partnership firm, it shall be signed by all the partners. In case of submission by one partner, the firm shall declare such partner as the lead partner through a declaration signed by all other partners. All correspondence shall thereafter be conducted in the name of the lead partner only.

5.4 If the bid is submitted by a company, it shall be signed by a duly authorised person holding power of attorney. A certified copy of the registered power of attorney shall accompany the bid.

5.5 The original bid shall be digitally signed by the person(s) duly authorised to sign on behalf of the Bidding firm. All pages of the bid, including any entries or amendments, shall be initialled by the person(s) signing the bid.

5.6 The uploaded certificates and financial documents shall be attested by a notary of the concerned court.

5.7 The bid shall ordinarily contain no alteration or addition, except those to comply with instructions issued by the Department, or as may be necessary to correct errors made by the Bidding firm, in which case the person(s) signing the bid shall initial such corrections.

5.8 The Agency/Bidder/Consultant shall bear all costs associated with the preparation and submission of the Bid, including the cost of presentations for clarifications, if any.

5.9 The Department in no case shall be held responsible or liable for any such costs regardless of the conduct or outcome of the bidding process.

5.10 Agencies/Bidders/Consultants are expected to undertake their visit for familiarization with site/office conditions.

5.11 All costs towards site/office visits, conferences, and submission of documents shall be borne by the Bidding firms themselves.

6. Department's Right to Accept Any Bid and to Reject Any or All Bids

6.1 The decision of the Departmental Tender Evaluation Committee, Water Resources Department, Government of Jharkhand regarding the opening of bids, evaluation, and acceptance of the bid shall be final and binding on all Bidding firms.

6.2 The Department reserves the right to accept or reject any bid, to annul the bid process, or to reject all bids at any time, without assigning any reason thereof.

7. Performance Guarantee

7.1 The selected Agency/Bidder/Consultant shall submit an irrevocable Performance Bank Guarantee, in the prescribed format (Annexure-2), from a Scheduled Commercial Bank, at the time of signing the Agreement.

7.2 The successful Agency/Bidder/Consultant shall provide a Bank Guarantee amounting to 5% of the Contract Value as Performance Guarantee for the Agreement period. An additional 5% shall be deducted from running account bills as security retention.

8. Signing of Agreement

8.1 On receipt of the Letter of Intent (LOI), the successful Agency/Bidder/Consultant shall report to the Office of the Executive Engineer, Water Ways Division, Ranchi, Water Resources Department, Government of Jharkhand for signing the Agreement in the prescribed form.

8.2 The signing of the Agreement shall take place only after the furnishing of the Performance Guarantee as per Clause 7 above. The Agreement shall be signed by the authorized person of the successful Agency/Bidder/Consultant within the specified period, failing which the LOI shall be withdrawn and the Bid Security/EMD shall be forfeited.

8.3 The successful Agency/Bidder/Consultant shall also sign a Non-Disclosure Agreement (NDA).

9. Time Schedule

The selected agency shall provide services within the timeframe finalized and mentioned in the Terms of Reference (TOR) and Scope of Work.

10. Information for Agency/Bidder/Consultant for Submitting the Bid

10.1 Each submitted bid shall be complete in all respects, including the proof of Tender Fee and Bid Security, which must accompany the submitted bid. The bid shall be enclosed with duly filled forms as prescribed.

10.2 The bid shall include a description of the firm/organization, the firm's general experience in the field of similar nature, the qualification and competency of the personnel proposed for the assignment, and the proposed work plan, methodology, and approach in response to the Terms of Reference (TOR). Necessary documents in support of technical capabilities in the annexed format (T-1, T-2, T-3 and T-4) must be submitted along with the proposal.

10.3 An affidavit swearing the correctness of technical data furnished by the Authorised Signatory must also be submitted in the annexed format (T-5).

10.4 The bids shall be annexed with Forms T-1, T-2, T-3, T-4 and T-5.

11. Technical Bid

While submitting the Technical Proposal, the Agency/Bidder/Consultant shall, in particular, ensure that:

- The prescribed Tender Fee and Bid Security are furnished;
- All forms are submitted in the prescribed formats and signed by the Authorised Signatory along with all relevant supporting documents;
- Power of Attorney is enclosed, if applicable;
- CVs of all Professional Personnel are included along with relevant supporting documents;
- Key Personnel are proposed only if they meet the Conditions of Eligibility laid down in this RFP;
- No alternative proposal for any Key Personnel is made and only one CV per position is furnished;
- CVs are recently signed and dated in blue ink by the respective Personnel and counter-signed by the Agency/Bidder/Consultant. Photocopied or unsigned/uncountersigned CVs shall be rejected.
- Professional Personnel proposed have good working knowledge of the English language;
- No Key Personnel has attained the age of 62 (sixty-two) years at the time of submitting the proposal;
- If any individual Key Personnel makes a false averment regarding qualification, experience, or other particulars, or fails to fulfil his/her availability commitment after signing of the Agreement, such personnel shall be liable to be debarred from future assignments of WRD for a period of 3 (three) years. The award of this Consultancy may also be liable to cancellation if a suitable replacement is not made available within 7 (seven) working days;
- The proposed team shall comprise experts and specialists ('Professional Personnel') in their respective areas of expertise, and managerial/support staff ('Support Personnel') such that the agency is able to complete the assignment within the specified time schedule;
- Bid shall remain valid for a period of not less than 120 (One Hundred Twenty) days from the last date of bid submission;
- Bid Security shall be valid for 45 (Forty-Five) days beyond the validity of the bid;
- The Agency/Bidder/Consultant shall submit PAN Card, Income Tax Returns of the last three (3) years, GST Registration Certificate and the latest GST Return Statement (Form 3B) under the Jharkhand Goods and Services Act, 2017;
- The Agency/Bidder/Consultant shall not be under Corporate Debt Restructuring (CDR) / Strategic Debt Restructuring (SDR) of the Bureau of Industrial and Financial Reconstruction (BIFR). A certificate from its Statutory Auditor declaring the same shall be submitted along with the bid;
- The successful Agency/Bidder/Consultant must submit the operation, maintenance, and management manual and schedule at the time of signing the Agreement; and

- The Agency/Bidder/Consultant shall be empanelled with any State or Central Government body for Design, Development, Implementation and Maintenance of Application Software and Websites.

11.2 Mandatory Key Personnel

The mandatory Key Personnel for the assignment include (but are not limited to):

- Senior Software Developer
- Software Developer
- Junior Software Developer
- Mobile Application Developer
- UI/UX Designer / Frontend Developer
- Database Administrator
- On-site Support / Technical Manpower

The agency may, in addition, procure the services of any other expert as per the requirements of a particular nature of work.

12. Opening of Bid

12.1 The amount, form, and validity of the Tender Fee and Bid Security furnished with each bid shall be checked. If the Bid Security does not conform to the amount and validity period as specified in the Notice Inviting Tender, the Agency/Bidder/Consultant's offer shall be declared invalid and the bid shall not be opened.

12.2 Subject to confirmation of the Bid Security by the issuing bank, bids accompanied with valid security shall be taken up for evaluation with respect to the Qualification Information and other information furnished in the Technical Part of the bid.

12.3 The Technical Bid shall be opened by the e-Procurement Officer at the place, date, and time specified in the Notice.

13. Evaluation of the Bid

The bid shall be evaluated through a two-bid system, comprising the Technical Bid and the Financial Bid. The bid shall be evaluated under the Quality cum Cost Based Selection (LQFB) method, with the following weightage:

Evaluation Segment	QCBS Weightage (%)
Technical Bid	85
Financial / Commercial Bid	15
Total	100

13.1 Evaluation of the Technical Bid

Technical evaluation shall be carried out on a 100-mark scale on the following criteria. Each scoring item shall be supported by appropriate documentary evidence (work orders, completion/ongoing certificates, audited turnover certificates with UDIN, ISO/empanelment certificates, deployment letters, etc.) failing which the relevant marks shall not be awarded.

Sl. No.	Short-Listing Criteria	Maximum Marks
1.1	Bidder's Average Annual Turnover for the last three (3) financial years (2022-23, 2023-24, 2024-25): - Up to INR 5 Crore : 2 marks > INR 5 Cr and ≤ INR 15 Cr : 3 marks > INR 15 Cr and ≤ INR 25 Cr : 4 marks > INR 25 Cr : 5 marks	5
1.2	Bidder/Agency project experience of similar nature in design, development and maintenance of software applications during the last two (2) financial years (2022-23, 2023-24, 2024-25). Documentary evidence — Completion / Ongoing Certificate. (Each project carries 2 marks; maximum 5 projects.)	10
1.3	Bidder is empanelled with any State Government Department, Central Government Ministry/Department, Public Sector Undertaking (PSU), or other Government body for the design, development, implementation, operation, and maintenance of application software and websites.	10
1.4	Certifications held by the Bidder (2.5 marks each): - ISO 9001:2015 (Quality Management System – QMS) - ISO 27001:2018 (Information Security Management System – ISMS) - ISO 20000-1 (IT Service Management System – ITSM) - Jharkhand valid Goods and Services Tax (GST) Registration Number	10
1.5	GMDIC Verified MSME — Government of Jharkhand	10
1.6	Experience in deployment of professional manpower resources with any State Government, Central Government, Public Sector Undertaking (PSU) or other Government organisation in a single order: - Between 1–5 manpower resources: 2.5 marks - Above 5 manpower resources: 5 marks	10
1.7	Key Personnel Qualification & Relevant Experience (2 marks each): Senior Software Developer 8+ years experience	10

Sl. No.	Short-Listing Criteria	Maximum Marks
	• Software Developer 5+ years experience • Junior Software Developer 2+ years experience • UI/UX Designer 4+ years experience • Mobile Application Developer 8+ years experience • Project Manager 12+ years experience	
2.1	1. Approach & Methodology, Understanding of Business Requirement — 3 marks 2. Solution, Technical & Business Reference Architecture Framework — 3 marks 3. Engagement Team & Work Plan — 4 marks 4. Project Plan with Work Breakdown Structure (WBS) and approach for Data Governance — 5 marks 5. Presentation — Understanding of the Scope of Work, User-friendly Application and Timeline for Deliverables — 20 marks	35
	TOTAL MARKS	100

Bidders securing a minimum of 80 (eighty) marks in the Technical Evaluation shall be considered technically qualified and eligible for opening of their Financial Proposals. However, in the event that fewer than three (3) bidders achieve the aforesaid threshold, the cut-off score may be reduced, but not below 70 (seventy) marks, to ensure that at least the top three (3) technically ranked bidders are shortlisted for financial evaluation.

13.2 Evaluation of the Financial Bid

(i) The agency shall submit the Financial Bid as per the Scope of Work for all five (5) Components A, B, C, D, and E. Information of technically qualified Agency/Bidder/Consultant shall be uploaded on the portal www.jharkhandtenders.gov.in.

(ii) If an Agency/Bidder/Consultant quotes NIL charges/consideration, the bid shall be treated as unresponsive and shall not be considered.

(iii) The Agency/Bidder/Consultant with the Lowest Quoted Financial Bid (LQFB) shall be awarded a 100% score for the Financial Bid.

(iv) Financial Scores for other than L1 Agency/Bidder/Consultant shall be evaluated using the following formula:

Financial Score (FS) = (Financial Bid of LQFB / Financial Bid of the Agency/Bidder/Consultant under evaluation) × 100

(v) The Financial Bid shall indicate total prices for all deliverables and services specified in this Bid Document, covering all five (5) project modules and the entire AMC period.

(vi) The rate quoted by the Agency/Bidder/Consultant shall be at the basic rate exclusive of all applicable taxes. During execution of the project, if the GST rate is changed by the GST Council, payment shall be made accordingly with the approval of the competent authority.

(vii) Any conditional bid shall be rejected.

(viii) The Total Score (TS) shall be obtained by weighting the Technical Score and Financial Score and adding them up. The proposal obtaining the highest combined technical and financial score shall be ranked H-1, followed by proposals securing lesser marks as H-2, H-3, etc. The Agency/Bidder/Consultant securing the highest combined marks shall be considered for contract award.

Total Score (TS) = (Technical Score × 0.85) + (Financial Score × 0.15)

(ix) Errors and Rectification: Arithmetical errors shall be rectified as follows: 'If there is a discrepancy between the unit price and the total price obtained by multiplying the unit price and quantity, the unit price shall prevail and the total price shall be corrected. If there is a discrepancy between words and figures, the amount in words shall prevail.'

14. Negotiations and Weight of Bid

14.1 The Bidder must be prepared to furnish the detailed cost break-up and other clarifications with respect to the proposal, as may be required to adjudge the reasonableness of the price proposal.

14.2 If negotiations with the H-1 Agency/Bidder/Consultant are successful, the award shall be made to that Bidder. If negotiations fail, the next-ranked Bidder shall be invited for negotiations. This process shall be repeated until an agreed contract is concluded.

14.3 The Water Resources Department, Government of Jharkhand is not bound to select any of the firms submitting proposals. As quality is the principal selection criterion, the Department does not bind itself in any way to select a firm bidding the lowest price.

14.4 Any manufacturing or construction firm with which a Bidder might be associated is not eligible to participate in any subsequent bid for goods or works resulting from or associated with the project of which this consulting assignment forms a part.

15. Substitution of Key Personnel

WRD expects all Key Personnel to be available during implementation of the Agreement. WRD shall not consider substitution of Key Personnel except for reasons of incapacity or health. Such substitution shall ordinarily be limited to Key Personnel, subject to equally or better qualified and experienced personnel being provided to the satisfaction of WRD.

16. Payment Schedule

Towards consideration for the role performed, the Agency shall be entitled to receive payment on the basis of the approved payment schedule. The payment schedule must be approved by the competent authority of the Department. The indicative phase-wise payment schedule is as follows:

S. No.	Phase / Milestone	Amount	Description
1	SRS, Inception Report, System Design Document and onsite deployment of the Development Team as per the Scope of Work for all five (5) modules.	30% of the Development Phase Value	Manpower deployment, SRS sign-off, SDD sign-off
2	Phase-1: Take-over of existing portals; development and Go-Live of (i) PPMS, (ii) Industrial Water Allocation Portal, and (iii) Contractor Registration & Empanelment Portal as per Scope of Work.	20% of the Development Phase Value	Phase-1 Go-Live Certificate
3	Phase-2: Development and Go-Live of (i) Water E-Tariff & Billing Portal and (ii) CMS-based Departmental Website as per Scope of Work.	10% of the Development Phase Value	Phase-2 Go-Live Certificate
4	Phase-3: UAT, Security Audit (CERT-In empanelled), System Integration Testing, Training of Master Trainers, Final Security Compliance and overall Go-Live of the Integrated Solution.	10% of the Development Phase Value	UAT Certificate, VAPT Compliance Certificate, Training Certificate, Final Go-Live Certificate
5	Phase-4: Quarterly AMC for 3 (Three) Years post Go-Live (10% per year, paid in equal quarterly instalments).	30% (10% per year)	Quarterly AMC against approved deliverables and SLA performance

All AMC payments shall be supported by attendance compliance certificates, SLA performance reports, and approved deliverables. AMC payment shall be released only after deduction of any liquidated damages, penalties, or recoveries due against the Agency.

17. Compliance with Labour Regulations

The Agency/Bidder/Consultant shall be bound to pay minimum wages as per labour laws, regulations, notifications, and bye-laws of the State Government, Central Government, or local authority, and any other labour law (including rules), regulations, or bye-laws that may be passed or notified in future, either by the State or Central Government or the local authority.

18. Hand Over and Take Over (HOTO) – Intellectual Property Rights and Source Code

18.1 Intellectual Property Rights (IPR)

18.1.1 Ownership

All Intellectual Property Rights (IPR) including but not limited to copyrights, patents, database rights, design rights, source code, and technical documentation developed under this Contract shall vest solely and exclusively with the Client (WRD) from the date of delivery or as otherwise agreed.

18.1.2 Licensing Responsibility

The Service Provider shall: (i) ensure that all third-party or open-source software components used are appropriately licensed; (ii) bear full responsibility for licensing compliance; and (iii) indemnify the Client against any legal, financial, or reputational consequences arising from unauthorised or non-compliant use of third-party or open-source software.

18.1.3 Open-Source Compliance

Only open-source components with permissive licences (e.g., MIT, Apache 2.0, BSD) may be used unless prior written approval is obtained from the Client. Copyleft licences such as GPL must be avoided unless explicitly permitted.

18.2 Source Code and Technical Artifacts

18.2.1 Source Code Handover

The complete and up-to-date source code (including modifications during implementation, testing, AMC, or security audit) shall be handed over in a secure digital format (e.g., encrypted USB, hard drive, or Git repository archive) and shall include source files, build scripts and libraries, configuration files, API and interface documentation, and deployment guides.

18.2.2 Ownership Declaration

The handed-over source code and all related deliverables shall become the exclusive property of the Client. The Service Provider shall retain no rights, access, or claims post-handover.

18.2.3 Backup and Recovery Documentation

A comprehensive backup and recovery plan shall be provided, covering periodic backup schedules, backup verification procedures, Disaster Recovery Plans (DRP), Recovery Point Objective (RPO) and Recovery Time Objective (RTO).

18.2.4 Version Control and CI/CD

The Service Provider shall utilise industry-standard version control systems (e.g., Git) and provide documentation on branching strategy, release tagging, and CI/CD pipeline setup and instructions.

18.3 Documentation Requirements

18.3.1 Post-Order Submission

Following the award of contract, the Service Provider shall submit a detailed implementation schedule with milestone breakdown, and technical datasheets of supplied items for approval.

18.3.2 Post-Commissioning Submission

After commissioning, the following documents shall be submitted: System Design Document (ER Diagrams, Database schema, UI/UX flow); User Manual (functionality, administration, troubleshooting); Backup and Recovery Document; Source Code Package; Warranty Certificates; Training Completion Report; and Change Logs and Version History.

18.4 Training Requirements

The Service Provider shall conduct comprehensive training on the delivered system, including system administration and configuration, metadata creation and data management, and functional usage of the application and mobile components. Training shall be conducted for at least five (5) Client personnel per module, for a duration of three (3) working days, at the Client's designated site or via secure remote platform. Customised training manuals (digital and printed), step-by-step guides, and training session recordings shall be provided.

18.5 HOTO Process

The HOTO process shall commence post final testing and commissioning and must be completed within fifteen (15) working days. The Service Provider shall submit a formal HOTO checklist including source code and deployment artifacts, final system and user documentation, admin and system credentials, hosting/cloud environment access, backup and disaster recovery documentation, training delivery report, and support and escalation contact matrix. The Client shall verify all deliverables and issue a Handover Acceptance Certificate upon successful completion.

18.6 Post-Handover Support

The Service Provider shall offer thirty (30) calendar days of free support post-handover to resolve minor bugs, configuration issues, and user queries. SLAs during this period shall be: Critical Issues – Response within 4 hours; Major Issues – Response within 8 hours; Minor Issues – Response within 24 hours.

18.7 Exit Management

In case of contract expiry or termination, the Service Provider shall ensure a seamless transition to the internal team or a new vendor; hand over all data, configurations, documentation, and access credentials; and provide assistance during the transition period, as mutually agreed. For high-availability or mission-critical applications, an escrow mechanism may be implemented for source code deposit and retrieval in case of vendor default.

19. General Provisions

19.1 The Agency/Bidder/Consultant shall make its own arrangements for the engagement of all Key Personnel/Positions, PMU, staff, local or other, and for their payment, housing, feeding, and transport.

19.2 Any legal litigation shall be settled in the judicial jurisdiction of Ranchi, Jharkhand, or the Supreme Court of India, New Delhi only.

SECTION-II: SPECIAL CONDITIONS

1. Public Acceptability of the Project

The Agency shall develop the proposed Integrated Software Solution in such a manner that it has high public acceptability and user-friendliness. The Agency shall make presentations before WRD on alternate possible architectural and design approaches to fulfil outlined objectives. The Agency shall make a comparative presentation of alternatives before the Client, clearly bringing out the comparative merits and demerits in terms of usability, accessibility, maintainability, public acceptability, and cost.

2. Delays in Performance

2.1 Timely implementation of the project shall be as per the provisions mentioned in the Agreement.

2.2 In case of delay in implementation of the project and/or any delay in performance during the contract period, the Agency shall be liable to any or all of the following actions:

- Imposition of Liquidated Damages.
- Forfeiture of Performance Guarantee.
- Termination of the Contract for default.

2.3 If at any time during commencement or performance of the contract the Agency faces difficulties impeding timely completion of the project and/or performance of services, the Agency shall promptly inform the Department in writing within 24 hours of such delay, including its cause and likely duration. As soon as practicable after receipt of such notice, the Department shall assess the situation and may at its discretion extend the time for commencement and/or performance with or without Liquidated Damages.

3. Liquidated Damages

3.1 In the event of failure by the Agency to implement the project as per the Agreement, the Department reserves the option to recover Liquidated Damages (and not by way of penalty) for late implementation as follows:

S. No.	Details of Delay	Liquidated Damages
(i)	Delay up to 25% of the implementation period	2% of the bid price
(ii)	Delay more than 25% and up to 50% of the implementation period	4% of the bid price
(iii)	Delay more than 50% and up to 75% of the implementation period	6% of the bid price
(iv)	Delay more than 75% and up to 100% of the implementation period	8% of the bid price
(v)	Delay more than 100% of the implementation period	10% of the bid price (maximum)

3.2 Such Liquidated Damages, if not paid by the Agency, shall be recoverable under the relevant provisions of the Public Damage Recovery Act, 1954 by the Department or its authorised competent authority.

3.3 Penalty for Technical Errors in Deliverables: If a technical error is found in any deliverable against any predetermined milestone causing delay in appraisal of any component of the project, the Agency shall be liable to pay Liquidated Damages as follows:

- If the Agency fails to comply with the first intimation of error within the prescribed time: 2% Liquidated Damages.
- If the Agency fails to comply with the second intimation: a further 3% Liquidated Damages.
- If the Agency fails to comply with the third intimation: a further 5% Liquidated Damages.
- If the deliverables are still not perfect after the fourth intimation, the Department reserves the right to terminate the Agreement, and no consultancy fee/payment/claim shall be admissible for the efforts of the Agency till the date of termination.

3.4 SLA-Based Penalties During AMC / Operations

During the AMC and operational phase, the following Service Level Agreements (SLAs) shall apply, with corresponding penalties for non-compliance:

SLA Parameter	Target	Penalty for Breach
Critical Incident Response Time	Within 2 hours (24x7)	Rs. 2,000 per incident per day delay
Critical Incident Resolution Time	Within 4 hours	Rs. 35,00 per incident per day delay beyond SLA
Major Incident Resolution	Within 8 hours	Rs. 5,000 per incident per day delay beyond SLA
Minor Incident Resolution	Within 24 hours	Rs. 1,000 per incident per day delay beyond SLA
Manpower Attendance	≥ 95% per month per resource	Pro-rata deduction for absent man-days.

4. No Claim Arrangements

The Agency shall not be entitled to make any claim whatsoever against the Department under, by virtue of, or arising out of this Contract, nor shall the Department entertain or consider any such claim. After the expiry of the Agreement, the Agency shall have to sign a 'No Claim' certificate in favour of the Department in such form as shall be required by the Department.

5. Quality Standards

5.1 Standards: The services provided under this Agreement shall conform to the standards mentioned in the Technical Specifications. Where no standards are mentioned, applicable Government of India / international standards shall apply.

5.2 Standard of Performance: The Agency shall carry out the services with due diligence and efficiency in accordance with generally accepted norms, techniques, and practices used in the industry. The Agency shall employ appropriate, advanced, secure, and effective technology, tools, and methodologies, including the latest IS Codes, OWASP, NIST, CERT-In and GIGW guidelines, and Government of Jharkhand IT Policies.

6. Confidentiality and Data Protection

6.1 The Agency shall treat the details of the Agreement as private and confidential, save in so far as may be necessary for the purposes thereof, and shall not publish or disclose the same in any trade or technical paper or elsewhere without the prior written consent of the Department.

6.2 The Agency shall not disclose, sell, or commercially exploit the data of WRD without the written permission of the Additional Chief Secretary / Principal Secretary / Secretary, Water Resources Department.

6.3 The Agency shall comply with the Information Technology Act, 2000 (and amendments), the Digital Personal Data Protection Act, 2023 (and Rules thereunder), and all applicable data protection laws.

7. Transfer of Rights

The Agency shall not transfer the Agreement or any part thereof to any other party without prior written approval of the Department.

8. Decisions and Interpretation

8.1 Except where otherwise specifically stated, the office designated by the Water Resources Department, Government of Jharkhand shall decide on any matter requiring interpretation. Any unresolved matter shall be referred to the Additional Chief Secretary / Principal Secretary / Secretary, Water Resources Department, whose decision shall be final and binding on both parties.

8.2 In case of dispute over the interpretation of any clause of the Agreement, the decision of the Additional Chief Secretary / Principal Secretary / Secretary, Water Resources Department shall be final and binding.

9. Suspension

9.1 The Department may, by written notice to the Agency, suspend the Agreement if the Agency fails to perform any of its obligations, including the carrying out of the services. Such notice shall: (i) specify the nature of the failure, and (ii) direct the Agency to rectify such failure within a specified period.

9.2 No payment shall be made to the Agency for such suspended period and no damage shall be claimed on account of such suspension.

10. Termination of the Contract

10.1 The Department shall have the right to cancel the Agreement if the Agency commits a breach of any condition. Breaches include, but are not limited to:

- Non-adherence to the time schedule of implementation.
- Stoppage of work without authorisation by WRD.
- Bankruptcy or liquidation of the Agency.
- Failure to correct defect/irregularity within reasonable time after notice.
- Failure to comply with instructions/orders of the Department.
- Failure to deliver any obligation within specified time or extension.
- Failure to perform any other obligation under the Agreement.
- Material breach of confidentiality, data protection, or security obligations.

10.2 Upon breach for any of the above reasons, the Department shall have the right to terminate the Agreement, forfeit the Security Deposit, and invoke the Performance Bank Guarantee.

11. Cessation of Rights and Obligations

Upon termination or expiry of this Agreement, all rights and obligations of the parties hereunder shall cease, except: (i) such rights and obligations as may have accrued on the date of termination or expiry; (ii) the obligation of confidentiality; and (iii) any right which a party may have under applicable law.

12. Cessation of Services Upon Termination

Upon termination of the Agreement, the Agency shall, immediately upon dispatch or receipt of such notice, take all necessary steps to bring the services to a close in a prompt and orderly manner, including handover of source code, data, documentation, and credentials.

13. Dispute Resolution

13.1 The Department and the Agency shall make every effort to resolve amicably, by direct negotiations, any disagreement or dispute arising under the Agreement.

13.2 If after 30 (thirty) days from commencement of such direct negotiations the dispute is not resolved, it shall be referred to the Additional Chief Secretary / Principal Secretary / Secretary, Water Resources Department, Jharkhand, whose decision shall be final and binding upon both parties.

13.3 Pending submission and/or decision on a dispute, the Agency shall continue to perform all its obligations under this Agreement without prejudice of final adjustment.

13.4 The Department may terminate this Agreement, by giving a written notice of minimum 30 (thirty) days, if the Agency fails to comply with any decision delivered.

14. Force Majeure

14.1 Notwithstanding the provisions of Clauses 12 and 13, the Agency shall not be liable for Liquidated Damages or termination for default if its delay in performance is the result of an event of Force Majeure. For the purpose of this Clause, 'Force Majeure' means an event beyond the control of the Agency and not involving any fault or negligence and not foreseeable. Such events may include, but are not restricted to, acts of the Department in its sovereign capacity, wars or revolution, fires, floods,

epidemics, quarantine restrictions, and freight embargoes. Force Majeure shall not include failure of electricity, printing system, or non-availability of raw material as a cause beyond control.

14.2 If a Force Majeure situation arises, the Agency shall promptly notify the Department in writing within 24 hours of such conditions and the cause thereof. The Agency shall continue to perform its obligations as far as is reasonably practical and shall seek all reasonable alternative means for performance not prevented by the Force Majeure event. The Department may terminate this Agreement, by giving written notice of minimum 30 (thirty) days, if the Agency is unable to perform a material portion of services for a period exceeding 60 (sixty) days due to Force Majeure.

15. Contract Validity and Extension

The contract shall be valid for three and a half (3.5) years and may be further extended by up to two (2) years based on the performance of the Agency, with a provision of up to 10% price escalation on mutually agreed terms, subject to the approval of the competent authority.

SECTION-III: TERMS OF REFERENCE (TOR) AND SCOPE OF WORK

1. Introduction

The Water Resources Department (WRD), Government of Jharkhand, hereby issues this Request for Proposal (RFP) to engage a qualified and experienced IT agency for the comprehensive modernisation and strengthening of the Department's digital ecosystem.

Through this composite RFP, the Department intends to undertake the design, development, re-engineering, implementation, secure hosting, security auditing, and long-term maintenance of its existing and proposed web applications and portals under a single integrated implementation framework.

The engagement shall cover five (5) interrelated components that collectively aim to enhance operational efficiency, transparency, security, and service delivery across the Department:

- **Component-A:** Project Progress Monitoring System (PPMS) — covering project monitoring, fund requisition, MIS, and dashboard for monitoring all projects of WRD.
- **Component-B:** Industrial Water Allocation Portal (Re-engineering & Modernisation of wrdd.jharkhand.gov.in) — covering applicant registration, water allocation/licensing, billing, JE-GRASS treasury integration, and multi-level approvals.
- **Component-C:** Contractor Registration & Empanelment Portal (Redesign & Redevelopment of wrdreg.jharkhand.gov.in) — covering contractor registration, renewal, payment integration, certificate management, and DigiLocker integration.
- **Component-D:** Water E-Tariff & Billing Portal — covering consumer registration, water-drawal entry, role-based JE→AE→EE workflow, tariff and billing engine, division-wise revenue accounting through JE-GRAS, and recovery management.
- **Component-E:** CMS-based Departmental Website — a modern, GIGW 3.0-compliant, bilingual, accessible, content-managed public-facing website for WRD with publication, RTI, citizen-services, and integration features.

At present, the Department operates multiple web-based applications supporting critical functions such as project monitoring, water allocation, contractor registration and empanelment, and financial management. While these systems have supported departmental operations, they require substantial upgradation to align with current technological standards and governance requirements. Key areas of improvement include modernisation of UI/UX, strengthening of cybersecurity, structured fund-requisition workflows, enhanced reporting and MIS, and seamless integration with other Government digital platforms.

The selected agency shall be responsible for end-to-end execution including development of new modules, complete re-development of existing applications, portal redesign and redevelopment, implementation of robust security architecture, hosting at the Jharkhand State Data Centre (JAPIT) / NIC Data Centre, and provision of sustained operational support along with deployment of on-site technical manpower.

The Department seeks to ensure that the proposed solution adheres to prevailing Government of India and Government of Jharkhand guidelines on security, interoperability, data protection, accessibility, and digital governance standards, thereby establishing a scalable, secure, and future-ready digital platform for the Water Resources Department.

2. Overall Objectives

The composite project aims to:

- Provide a unified, secure, and citizen-centric digital ecosystem for WRD.
- Streamline internal departmental workflows, including project monitoring, fund requisition, contractor empanelment, water allocation, and tariff/billing management.
- Strengthen operational transparency through real-time reporting, MIS, and BI dashboards.
- Ensure application availability, security, accessibility, and regulatory compliance.
- Deploy skilled manpower for training, support, and day-to-day operations.
- Enable seamless integration with State e-Governance platforms (Single Window Clearance System, JE-GRAS, DigiLocker, SMS Gateway, e-Mail services, GIS, NIC Data Centre).
- Establish a modern, GIGW-compliant, multilingual public-facing website for the Department.
- Maximise revenue realisation through automated tariff calculation, division-wise revenue credit, and efficient recovery management.

3. Inadequacies in the Current Software / Modifications Required

The existing systems of WRD lack the following functionalities, which the proposed Integrated Solution must address:

- Absence of an automated, end-to-end Fund Requisition Module integrated with project metadata.
- Lack of real-time SMS notifications and OTP-based authentication.
- Limited reporting capabilities and absence of a BI-enabled MIS dashboard.
- No digital, role-based workflow for water-tariff billing, JE→AE→EE approvals, and division-level revenue accounting.
- Absence of a dedicated Water E-Tariff & Billing System integrated with JE-GRAS for division-wise revenue credit.
- Absence of a modern, GIGW 3.0-compliant CMS-based Departmental Website with bilingual and accessibility features.
- Inadequate security compliance, including non-compliance with CERT-In VAPT, OWASP Top 10, and SDC hosting security mandates.
- Lack of structured Operations & Maintenance support and a dedicated Project Management Unit (PMU).
- Inadequate Google Maps / GIS integration for project geo-visualisation.
- Limited Disaster Recovery, Business Continuity, and Backup arrangements.

4. Overall Scope of Work — Composite Project

The composite scope of work covers the following macro-deliverables across all five (5) components:

12. Conducting a detailed As-Is study of the existing portals (PPMS, wrdd.jharkhand.gov.in, wrdreg.jharkhand.gov.in, Water E-Tariff and Billing portal, and the existing departmental CMS based website).
13. Preparation and sign-off of Software Requirement Specification (SRS), System Design Document (SDD), and UI/UX prototypes for each component.
14. Take-over and handover of existing applications, code, data, and infrastructure from the existing vendors / NIC.
15. Re-engineering, redesign, and redevelopment of existing portals and websites using a modern technology stack.
16. Development of new modules and the new Water E-Tariff & Billing Portal.
17. Development of a new CMS-based Departmental Website.
18. Complete and accurate Data Migration from the legacy systems including data cleansing, deduplication, validation, and parallel run for not less than thirty (30) days.
19. Integration with Single Window Clearance System (SWCS), JE-GRAS, DigiLocker, SMS Gateway, Email Gateway, Google Maps API, and GIS systems as applicable.
20. Implementation of OWASP Top 10 compliance, AES-256 encryption at rest, TLS 1.3 in transit, MFA, RBAC, WAF, and DDoS protection.
21. Conducting CERT-In empanelled VAPT before Go-Live and annually thereafter.
22. Hosting at the Jharkhand State Data Centre (JAP-IT) / NIC Data Centre with HA, DR, daily/weekly backups, and 90-day retention.
23. Training of departmental officers and master trainers along with delivery of comprehensive bilingual user manuals (Hindi & English).
24. Deployment of on-site Project Management Unit (PMU) manpower for the entire AMC duration of three (3) years post Go-Live.
25. Provision of three (3) years of comprehensive Operation & Maintenance support, including bug-fixing, version upgrades, OS/DB/browser compatibility updates, change-request handling, and reinstallation if needed.
26. Annual VAPT, source-code review, and remediation as per CERT-In advisories.
27. Periodic submission of MIS reports, SLA performance reports, and quarterly health reports to WRD.
28. Execution of any other related work directed by the Department from time to time, including configuration, integration, custom reporting, and minor enhancements within the scope of AMC.

5. Component-A: Project Progress Monitoring System (PPMS)

5.1 Objective

The objective of Component-A is to enable seamless and secure project monitoring and fund requisition through a dedicated web and mobile application. This system shall integrate with Google APIs, SMS Gateway, and OTP-based authentication, provide real-time MIS dashboards, and ensure long-term operational stability through timely updates, backups, and trained manpower support.

5.2 Scope of Work — Component-A (PPMS)

The proposed web and mobile application shall digitise and automate the project fund requisition process, enhance reporting and MIS, ensure data security, and facilitate long-term operational stability. The activities covered include:

- Development of a Fund Requisition Module with hierarchical approval workflows (creation, escalation, finance review, sanction, release).
- Enhancement of the Reporting Module with multi-dimensional filters, multi-format exports, and analytics.
- Integration with Google Maps API for dynamic project location visualisation, geotagging, route mapping, and satellite imagery — including secure API key management, compliance with Google's terms of use, and procurement of necessary licences (cost to be included in the financial bid).
- Integration with any other Government software/tool as directed by the Competent Authority.
- Integration with SMS Gateway for alerts and OTP-based authentication.
- Security audit compliance through a CERT-In empanelled agency, including pre-Go-Live audit and periodic re-audits during AMC.
- Application hosting support at the Jharkhand State Data Centre (JAP-IT) / NIC Data Centre, including deployment, configuration, and coordination with hosting authorities.
- Deployment of on-site technical manpower as per requirements of the Department.
- Training and hand-holding of departmental staff and officers.
- AMC services including version upgrades, patching, data updates, change requests handling, and reinstallation as needed.
- As per JAP-IT mandate, only security-audited applications are permitted to be hosted. After hosting, periodic security audit shall be performed and the Agency shall patch all audit objections within the timelines stipulated by JAP-IT.
- Execution of any other related work directed by the Department.
- All necessary server access, VPN connectivity, and other required system access permissions shall be provisioned by the Department to the selected Bidder/Agency/Consultant to facilitate the smooth operation, maintenance, and upgradation of the system throughout the duration of the contract.

5.3 Project Initiation

- Conduct an initial kick-off meeting to discuss project objectives, expectations, and timelines.
- Establish clear lines of communication and designate key points of contact.
- Submit Project Charter and Detailed Work Plan within 7 days of issue of Letter of Acceptance.

5.4 Requirements Analysis

- Collaborate with the Department to gather detailed requirements for the online platform.
- Conduct meetings with stakeholders at the Engineer-in-Chief, Chief Engineer, Superintending Engineer, Executive Engineer, and Junior Engineer levels to identify specific needs and functionalities.
- Prepare and obtain sign-off on the Software Requirements Specification (SRS) Document, including functional and non-functional requirements, use-cases, business rules, and acceptance criteria.

5.5 System Design and Architecture

- Develop a detailed System Design Document (SDD) including High-Level Design (HLD), Low-Level Design (LLD), database schema, API specifications, and deployment architecture.
- Ensure scalability, security, and compatibility with existing State Government and Central Government systems.
- Present the design for review, feedback, and formal sign-off by WRD.

5.6 Development

- Develop the online platform incorporating the approved design and requirements.
- Provide regular fortnightly progress updates and conduct milestone reviews.
- Implement features for project management, task tracking, milestone monitoring, and inter-department coordination.
- Maintain a Git-based version control with branching strategy and CI/CD pipelines.

5.7 Testing

- Conduct thorough unit testing, integration testing, system testing, performance testing, and security testing.
- Perform User Acceptance Testing (UAT) with WRD stakeholders to ensure alignment with expectations.
- Address and resolve all feedback from each testing phase, with documented test results and reports.

5.8 Deployment

- Deploy the online platform in a controlled staging environment for a pilot/UAT phase.
- Monitor system performance and gather user feedback during the pilot phase.
- Make necessary adjustments and optimizations based on user feedback.
- Migrate to production at JAP-IT / NIC Data Centre upon WRD approval, post Security Audit clearance.

6. Modules for the PPMS Portal

6.1 Module A — Fund Requisition Module

The Fund Requisition Module shall streamline the financial demand lifecycle for projects under various schemes and jurisdictions. It shall enable departmental officials to initiate, manage, and track financial demands for new or ongoing projects, and ensure transparent coordination with the Finance Department and administrative hierarchy. The module shall be designed with security, flexibility, and accuracy in handling financial data.

6.1.1 Raise / Construct a Fund Requisition Against a Project

- Allow users (officials) to initiate a financial fund requisition for a project under a specific scheme.
- Fund requisition form fields shall include: Project Name/ID, Scheme Name, Sub-Head of Scheme, Head of Account, Financial Year, Amount Requested, Justification for Demand, Supporting Documents Upload, Jurisdiction/Location, and Timeline (Expected project start/end dates).
- Auto-population of fields from project metadata wherever applicable.
- Save-as-Draft and Submit options.
- Auto-generated unique Fund Requisition Number per project/scheme/year.

6.1.2 Edit / Update Fund Requisition

- Allow fund-requisition creators to edit/update demand details before submission to higher management.
- Edit requests shall be version-controlled and logged.
- Updates permissible only in pre-submission or post-rejection states.
- Validation checks to prevent conflicting or duplicate entries.

6.1.3 Delete Fund Requisition

- Permissible only in 'Draft' status and before submission.
- Soft delete with audit logging; admin-controlled hard delete in exceptional cases.

6.1.4 Send Fund Requisition Request to Higher Management

- Hierarchical escalation to higher authority post creation.
- Notification/alert to approver with full demand details.
- Demand status changes to 'Pending Review'.
- Reviewer interface to Accept / Reject / Send Back for Revision, with mandatory remarks for Reject or Send Back actions.

6.1.5 Search Fund Requisition via Scheme and Head of Account

- Search functionality on the review/approval dashboard for higher management, finance, and admin.
- Filters: Scheme Name, Head of Account, Status (Draft, Pending, Accepted, Rejected, Under Finance Review, Released), Date Range, Project ID/Name.
- Option to export results to Excel/PDF.

6.1.6 Processed Stages After Acceptance

- On approval by the reviewing officer, fund requisition moves to 'Under Finance Review'.

- Finance officials receive notification and access to review submitted fund requisitions.
- Finance interface to Accept / Reject / Request Modification.
- Editable fields by Finance: Allocated Amount, Release Date, Fund Code.
- Status updated to: Approved by Finance / Rejected by Finance (with reasons) / Pending Clarification.

6.1.7 Release of Requisite Fund from Admin End

- Once accepted by Finance, Admin is notified for release.
- Admin Panel actions: View all approved fund requisitions; Input Fund Release Reference Number; Upload Sanction/Release Document; Mark as Released; Generate downloadable Fund Release Certificate (PDF).
- Status updated to 'Released' and visible in dashboard.

6.2 Module B — Reporting Management Module

- Customisable Reports: Reports based on dates, transaction status (success, failed, pending), payment modes, arrears, or any other key data points.
- Report Formats: Export in PDF, Word, Excel, CSV, etc.
- Scheduled Reports: Automatically generated reports to stakeholders on daily, weekly, or monthly basis.
- Sample Reports: Monthly Progress Report, Pending Work Report, Level-wise Progress and Pendency Report.
- Auto-Generated MIS Sheets: At the start of each month, automatic generation of an MIS Report containing physical and financial progress and current status.
- Downloadable in PDF, Word, Excel, CSV formats.
- Robust Search, Filtering, and Downloading: Search by project, status, timeline, user, level, or date; filter by status, value, date range, progress percentage.

6.3 Module C — Customization of MIS & Dashboard

The MIS & Dashboard Module shall serve as the central hub for monitoring and managing the various processes within the project. This module shall consolidate data from other modules, providing a comprehensive view of system performance. The MIS shall ensure that all relevant data is captured, processed, and stored systematically, allowing for real-time reporting and analysis. The dashboard shall offer visual representations of key metrics such as the number of applications processed, grievances resolved, and funds disbursed through the system.

The dashboard shall be visually intuitive and data-driven, providing real-time insights and analytics. It shall be tailored to the specific needs of WRD, incorporating Key Performance Indicators (KPIs), metrics, and data visualisations that enable swift and informed decision-making. The dashboard shall be user-friendly with customisable views catering to different roles within the project — from field officers to senior administrators. Alerts and notifications shall be configurable to highlight critical issues.

6.4 Module D — BI-Enabled Dashboard

The proposed MIS & Dashboard shall include BI-enabled features and analytics capabilities.

6.4.1 Key Metrics and KPIs

- Financial Metrics: Monthly disbursement totals, pending payments, arrears, failed transactions.
- Beneficiary Metrics: Active beneficiaries, new enrolments, deactivations, error logs.
- Performance Metrics: System uptime, transaction processing time, data validation success rate.
- Customised KPIs that users can set and track based on operational needs.

6.4.2 Data Visualisation Tools

- Interactive Charts: Dynamic pie charts, bar graphs, line charts, and tables for financial transactions, user activity, and system performance.
- Geographical Mapping: GIS tools to display beneficiaries and disbursements across regions.
- Drill-Down Capabilities: Drill from high-level data into detailed views (individual beneficiary, specific region, etc.).

9. Component-B: Contractor Registration & Empanelment Portal — Redesign & Redevelopment (wrereg.jharkhand.gov.in)

9.1 Objective

The Department proposes complete redesign, redevelopment, implementation, hosting, deployment, and long-term maintenance of the WRD Contractor Registration & Empanelment Portal. The existing Version 3.0 portal shall be comprehensively overhauled using a modern technology stack while retaining existing functionalities and introducing significant enhancements.

9.2 Portal Redesign & UI/UX Overhaul

- Full compliance with GIGW 3.0 guidelines issued by NIC / MeitY.
- Fully responsive design supporting Desktop, Tablet, and Mobile devices.
- WCAG 2.1 Level AA accessibility compliance including screen-reader compatibility, keyboard navigation, high-contrast mode, and adjustable font size.
- Bilingual interface (Hindi & English) with toggle functionality across all modules.
- Adherence to Government of Jharkhand branding guidelines.
- Intuitive navigation with breadcrumbs, contextual tooltips, and guided workflows.
- Professional dashboards with interactive charts and real-time data visualisation.
- Dark mode support, customisable themes, and Progressive Web App (PWA) capabilities for mobile users.

9.3 Public Portal Modules

9.3.1 Module A — Home Page & Information Module

- Dynamic CMS-driven home page with configurable banners, sliders, and announcements.
- Departmental information, Secretary's message, organisational structure, and contact directory.
- SOP section with downloadable documents.

- Latest news ticker with archive facility.
- FAQ section, Sitemap, Terms of Use, Privacy Policy, Accessibility Statement.
- Grievance Redressal module with ticket-based tracking.

9.3.2 Module B — Contractor Registration Module (New Registration)

- Online self-registration with Aadhaar-based e-KYC and OTP verification.
- Multi-step guided registration wizard with validation and progress indicators.
- Upload facility for required documents including Photograph, Signature, PAN Card, Address Proof, Affidavit, Incorporation Certificate, MOA/AOA, Partnership Deed, etc.
- Auto-population from PAN/Aadhaar databases wherever permitted.
- Digital signature support.
- Draft save functionality.
- Acknowledgement number generation with SMS/Email notifications.
- Class-wise registration (Class 1–4) with eligibility validation.

9.3.3 Module C — Contractor Renewal Module

- Online renewal with pre-filled data.
- Document re-upload facility.
- Renewal fee calculation and payment via E-GRAS.
- Renewal status tracking with SMS/Email alerts.

9.3.4 Module D — Payment Integration Module

- Seamless E-GRAS integration supporting Net Banking, Debit Card, Credit Card, UPI.
- Auto-reconciliation with challan verification.
- Offline payment recording facility.
- Payment receipt generation.
- Refund management for rejected or cancelled applications.

9.3.5 Module E — Applicant Dashboard

A personalised dashboard shall display application status, certificate validity, renewal alerts, and payment history. It shall provide download facility for Registration Certificate, Payment Receipts, and Acknowledgements; a notification centre; profile management; grievance submission and tracking; and work-order register access.

9.4 Department Admin Panel Modules

9.4.1 Module A — Multi-Level Dashboard & Analytics

- Role-based dashboards for ASO, SO, US, DS, JS, and EIC.
- Real-time data summaries with graphical analytics and KPI tracking.
- Date-range filters and export functionality.

9.4.2 Module B — Application Processing Workflow

ASO → SO → US → DS → JS → EIC

Features shall include:

- Segregation of New and Renewal cases.
- Advanced Inbox with search and filters.
- Application view with document viewer (zoom, rotate, download).
- Verification checklist and bilingual process interface.
- Bulk action capability.
- Auto-escalation for SLA breaches.
- Complete action audit trail.

9.4.3 Module C — Certificate Management Module

- Digital certificate generation with auto-population.
- QR Code-based verification for authenticity.
- Certificate pending queue with challan verification and OTP-based approval.
- Send Reminder functionality.
- Certificate revocation and cancellation workflow.
- DigiLocker integration for pushing issued certificates.

9.4.4 Module D — Comprehensive Reporting Module

- Application reports with multi-criteria search.
- Revenue reports displaying total revenue.
- Pendency reports with role-wise distribution.
- Disposal reports with date-range filters.
- Custom report builder for ad-hoc reports.
- Scheduled report generation with automated email delivery.

9.5 Additional Modules

- SMS & Email Notification Engine with automated alerts at every stage.
- Audit Trail & Logging covering all user actions, file movements, and system events.
- Document Management System (DMS) with centralised repository, version control, and search capability.
- Grievance Redressal Module with SLA-based resolution and escalation matrix.
- Blacklist Management module with public display capability.
- Master Data Management module for managing dropdown values, fee structures, classifications, and configurable parameters.
- User Management module for creation, modification, and deactivation of departmental users with role-based permissions.
- API Gateway providing a secure RESTful API layer for future integration with departmental systems and third-party platforms.

8. Component-C: Industrial Water Allocation Portal — Re-engineering & Modernisation (wrdd.jharkhand.gov.in)

8.1 Objective

The Water Resources Department (WRD), Government of Jharkhand, proposes comprehensive re-engineering and modernisation of its existing Industrial Water Allocation Application. The objective is to transform the current system into a modern, scalable, secure, responsive, and citizen-centric e-Governance platform aligned with contemporary technology and governance standards.

The selected agency shall execute this assignment under the supervision of the Department and shall coordinate with Single Window Jharkhand and JE-GRASS officials wherever required, while maintaining independent technical responsibility for solution delivery.

8.2 Core Functional Modules

8.2.1 Registration & Applicant Management

- Applicant onboarding with structured organisation profile management.
- Aadhaar / PAN / GST validation with secure document upload and verification mechanisms.
- Integration with the Single Window Clearance System (SWCS).
- Unique Application ID generation for each submitted request.

8.2.2 Water Allocation & Licensing

- Selection of water source (river / canal / reservoir) with quantity allocation.
- Seasonal validation and policy-driven allocation engine.
- Automated licence generation through structured digital workflows.

8.2.3 Renewal & Annual Water Demand

- Filing of annual returns and renewal application processing.
- Automated reminders for due renewals and validation of outstanding dues.

8.2.4 Billing & Revenue Module

- Water tariff computation based on usage parameters.
- Automated late penalty calculation.
- Invoice generation and issuance of demand notices.

8.2.5 Secure Treasury Payment Integration (JE-GRASS)

- Challan generation and secure payment verification.
- Receipt generation and treasury reconciliation.
- Handling of failed transactions with complete audit trail and status tracking.

8.2.6 Multi-Level Approval Workflow System

A structured digital workflow shall be implemented comprising:

AE → EE → SE → CE → EIC → Secretary

The system shall support digital scrutiny and file movement tracking, entry of remarks and objections, Hold/Reject/Approve functionality, and complete visibility of processing status at each stage.

8.2.7 Modification & Compliance

- Change-request processing.
- Department-initiated reduction of allocation.
- Surrender processing.
- Violation recording and compliance handling.

8.2.8 Inspection & Enforcement

- Field inspection data entry and record maintenance.
- Violation recording.
- Issuance of show-cause notices.
- Compliance monitoring and enforcement tracking.

8.2.9 Dashboard & Public Tracking

- Citizen tracking using Application Number.
- Officer dashboard displaying pending approvals, revenue collection status, and expiring licences.

8.2.10 Reports & MIS

- Registration statistics.
- Allocation and renewal data.
- Revenue collection reports.
- District-wise and source-wise usage.
- Pending cases tracking.
- API communication logs.

8.2.11 Functional Requirements – Contractor Registration & Revenue Management Reporting

The system shall provide a robust reporting and analytics module enabling Financial Year (FY)–wise and contractor class–wise segregation of registered contractors. The application must support dynamic filtering capabilities to generate reports on:

- Number of contractors registered in each financial year.
- Classification-wise contractor distribution (as per defined categories/classes).
- Total revenue collected from contractor registrations, categorised by FY and contractor class.

The module shall allow users to apply multiple filters and export reports in standard formats (Excel/PDF) for administrative and audit purposes.

8.2.12 GST Integration with Document Upload Facility

The application shall incorporate a mandatory GST field within the contractor registration process. The selected agency shall ensure the following:

- Provision for capturing GST Number as part of contractor registration details.

- Only Jharkhand State-registered GSTIN shall be considered valid for registration and empanelment.
- Facility for uploading relevant GST registration documents as supporting attachments.
- Validation mechanism for GST number format (as per applicable standards).

Additionally, the system must provide a mechanism to update and onboard GST details for already registered contractors. This may include:

- Prompting existing contractors to update GST details upon login.
- Administrative provision to update or verify GST details in legacy records.
- Ensuring that GST information is uniformly maintained across all contractor records.

8.3 Integration Requirements

- Single Window Clearance System (SWCS) for inter-departmental communication.
- JE-GRASS Treasury for secure payment processing and reconciliation.
- SMS Gateway and Email Server for alerts and notifications.
- API-first architecture ensuring future mobile application readiness.
- GIS-based map integration (future-ready architecture) for geographical visualisation.

8.4 Report Modules

- Registration Report Module with detailed analytics.
- Renewal Report Module tracking renewal statistics.
- JE-GRASS Payment Report Module for transaction-level visibility.
- API Request & Response Report summarising all system communications.
- Approval Staging Report tracking approvals, rejections, and holds across workflow levels.
- Application Details Report and Application Summary Report for individual and aggregate views.

10. Component-D: Water E-Tariff & Billing Portal

10.1 Project Background and Objective

The Water Resources Department (WRD), Government of Jharkhand, proposes to implement a comprehensive Water E-Tariff Management System for digitisation of the complete process relating to water allocation, consumption assessment, tariff calculation, billing, approval, collection, accounting, and monitoring of water-usage charges from various consumer categories.

The objective of the proposed system is to:

- Digitise the complete billing and collection process.
- Ensure transparent tariff calculation.
- Improve revenue realisation.
- Strengthen monitoring and audit controls.
- Eliminate manual errors.
- Enable division-wise accounting.
- Provide citizen-centric online payment services.
- Ensure compliance with Government financial rules.
- Maintain complete digital records for future reference.

The application shall be accessible at: Head Office; Circle Office; Division Office; Sub-Division Office; Consumer-end Portal.

The entire system shall be developed as a secure web-based application compliant with GIGW 2.0 / 3.0, Government of Jharkhand IT policies, cyber-security standards, accessibility standards, and treasury integration norms.

10.2 Scope of Work — Component-D

The selected Bidder shall be responsible for:

- Requirement gathering and process study.
- Application design and software development.
- User-role configuration and workflow automation.
- API integration with JE-GRAS, treasury, banks, and UPI services.
- Testing, deployment, training, and security audit.
- Hosting support at JAP-IT / NIC Data Centre.
- Annual Maintenance Support for three (3) years post Go-Live.

10.3 Functional Modules

10.3.1 Consumer Registration Module

The system shall provide online registration of consumers.

Consumer Categories: The software shall support the following categories:

- Government Departments

- Public Sector Undertakings
- Municipal Bodies
- Private Companies
- Industrial Units
- Commercial Units
- Institutions
- Individual Consumers

Registration Details: The system shall capture the following information for each consumer:

- Consumer Name and Organisation Name
- Address and Contact Details
- GST Number and PAN
- Water Source and Division Mapping
- Category of Usage
- Agreement Details and Allocation Quantity
- Sanction Order Reference and Validity Period

Each consumer shall be assigned: Unique Consumer ID; Login Credentials; Billing Account Reference.

10.3.2 Water Drawal Entry Module

The Junior Engineer (JE) shall enter:

- Meter Reading
- Manual Drawal Quantity
- Period of Consumption
- Previous Reading and Current Reading
- Total Consumption
- Excess Drawal
- Unauthorised Usage

The system shall validate consumption data, detect abnormal consumption, compare historical usage, and flag suspicious entries for review.

10.4 Role-Based Access and Workflow System

The software shall include a role-based login and approval system, with the following roles and responsibilities:

10.4.1 Junior Engineer (JE)

JE shall:

- Enter water consumption data.
- Prepare provisional bills.
- Apply tariff.

- Add surcharge and penalty.
- Upload supporting documents.
- Submit bill for verification.

JE shall NOT have authority to approve bills.

10.4.2 Assistant Engineer (AE)

AE shall:

- Verify consumption entries.
- Validate tariff applied.
- Verify penalties.
- Review consumer category.
- Check previous dues.
- Return to JE for correction if required (and vice versa).
- Forward verified bill to Executive Engineer.

AE may record remarks, seek clarification, and maintain verification history.

10.4.3 Executive Engineer (EE)

Executive Engineer shall:

- Review verified bill.
- Approve final demand.
- Generate demand notice.
- Authorise payment link.
- Monitor revenue.
- View division collection.
- Approve recovery actions.

10.4.4 Consumer Login

Consumers shall be able to:

- View demand notice.
- Download bill.
- View payment history.
- Receive payment link.
- Make online payment.
- Download receipt.
- Raise grievance.
- Track application status.

10.5 Tariff and Billing Module

The billing engine shall support the following tariff components:

- Fixed charges.
- Variable charges.
- Category-wise rates.
- Seasonal rates.
- Slum / industrial differential.
- Excess usage charges.
- Delayed payment penalty.
- Interest on arrears.

The system shall auto-calculate charges, penalties, and applicable taxes (GST), and generate digital demand notices. Only authorised officers shall have privileges to revise tariff rates, with full audit trail.

10.6 Approval Workflow

The software shall support the following workflow:

JE → AE → Executive Engineer → Consumer

Workflow features shall include:

- Maker-checker system.
- Return for correction at any stage.
- Time-stamping of every action.
- Digital remarks.
- Configurable escalation matrix.
- Pending alerts and dashboards.
- Comprehensive audit logs.

10.7 Payment Integration Module

The selected Bidder shall integrate the system with the following:

10.7.1 Mandatory Integrations

- JE-GRAS Jharkhand.
- Treasury systems.
- Bank gateways.
- UPI services.

Payment shall be made through: JE-GRAS, Net Banking, UPI, Debit Card, Credit Card, NEFT/RTGS.

10.7.2 Division-wise Revenue Credit

A key requirement of the system is that each payment made by the consumer shall be credited directly into the respective Division-level designated bank account of the Water Resources Department.

The software shall:

- Map consumer to division.

- Identify division account.
- Route payment correctly to the mapped division account.
- Generate receipt division-wise.
- Reconcile payments automatically.

10.7.3 Single Window Payment System

The software shall provide:

- Single payment page.
- Multiple payment channels.
- Auto account mapping.
- Instant payment confirmation.
- Auto receipt generation.

This shall ensure ease for consumers, faster payment, transparent accounting, and better monitoring.

10.8 Revenue and Recovery Management

The system shall support:

- Current demand.
- Arrear demand.
- Penalty demand.
- Defaulter list management.
- Recovery notices generation.
- Payment reconciliation.
- Refund handling.
- Comprehensive revenue reports.

10.9 Dashboard and MIS

10.9.1 Administrative Dashboard

- Consumer count.
- Revenue collected.
- Pending bills.
- Pending approvals.
- Division performance.
- Outstanding dues.
- Defaulters.

10.9.2 Reports

- Division-wise revenue.
- Consumer-wise billing.
- Payment status.

- Monthly collection.
- Penalty report.
- Audit trail.
- Outstanding report.

10.10 Security Requirements (Component-D Specific)

The application shall provide:

- Role-based access control (RBAC).
- Two-factor authentication (2FA / OTP).
- SSL certificate (TLS 1.3).
- Encrypted database (AES-256 at rest).
- Strong password policy.
- Session management with idle-timeout.
- User log tracking.
- API security with token-based authentication.
- Backup encryption.

10.11 GIGW Compliance

The software shall comply with:

- Accessibility guidelines (WCAG 2.1 Level AA).
- Mobile responsiveness.
- Standard navigation.
- Metadata.
- Search optimisation.
- Performance optimisation.
- Security compliance.
- Bilingual interface (Hindi & English) where required.

10.12 Security Audit

The Bidder shall conduct, through a CERT-In empanelled agency:

- Annual VAPT.
- Source code audit.
- Penetration testing.
- Compliance audit.
- Risk mitigation and patch closure.

Frequency: Once every year for the entire 3-year AMC duration.

10.13 Support and Maintenance (Component-D)

The Bidder shall provide:

- Corrective maintenance.
- Preventive maintenance.
- API monitoring.
- Payment support.
- Bug fixing.
- Performance tuning.
- Security patching.

Duration: Three (3) years after Go-Live.

10.14 Training (Component-D)

Training shall be conducted for:

- Junior Engineers (JE).
- Assistant Engineers (AE).
- Executive Engineers (EE).
- System Administrators.
- Finance staff.

Training materials shall include: User manual, Process manual, Video tutorials, and Helpdesk support.

10.15 Deliverables (Component-D)

- SRS document.
- Design document.
- Source code.
- API documentation.
- Security reports.
- User manuals.
- Training documents.
- Test reports.
- Go-Live report.

10.16 Ownership

- All data shall remain the property of WRD.
- Payment records shall remain Government property.
- Consumer records shall remain Department property.
- Confidentiality is mandatory throughout and after the contract period.

10.17 Implementation Timeline (Component-D)

Activity	Timeline
Requirement Analysis	30 Days
Development	60 Days
API Integration	20 Days
Testing	20 Days
Training	10 Days
Go-Live	10 Days

Total project duration for Component-D: Approximately 4–5 months from sign-off of SRS, to be aligned within the overall 6-month integrated development window.

10.18 Expected Benefits

The system shall provide:

- Transparent billing.
- Faster revenue collection.
- Reduced manual work.
- Accurate tariff application.
- Better monitoring.
- Improved accountability.
- Division-level accounting.
- Consumer convenience.
- Compliance with Government norms.

11. Component-E: CMS-based Departmental Website

11.1 Project Background and Objective

The Water Resources Department (WRD), Government of Jharkhand, currently operates a public-facing departmental website that serves as the primary digital interface for citizens, contractors, industries, institutional consumers, employees, and other stakeholders. The existing website lacks modern Content Management System (CMS) capabilities, full GIGW 3.0 compliance, robust accessibility features, true bilingual support, and structured integration with departmental applications such as PPMS, the Industrial Water Allocation Portal, the Contractor Registration Portal, and the proposed Water E-Tariff & Billing Portal.

Through this composite RFP, the Department intends to undertake complete redesign and redevelopment of its public-facing departmental website on a modern, secure, scalable, open-source CMS platform. The new website shall serve as the unified citizen-engagement and information-dissemination platform of WRD and shall be tightly integrated with the four (4) other components developed under this composite project.

The objective of Component-E is to:

- Provide a single, authoritative, citizen-centric web presence for WRD.
- Ensure full compliance with the Guidelines for Indian Government Websites (GIGW 3.0) issued by NIC / MeitY.
- Achieve WCAG 2.1 Level AA accessibility compliance.
- Deliver a fully bilingual (Hindi & English) experience with toggle support across all pages, content types, and dynamic data.
- Empower departmental staff to publish and manage content directly through a user-friendly CMS, without dependence on the IT vendor.
- Provide deep linking and Single Sign-On (SSO) integration with the four (4) other components — PPMS, Industrial Water Allocation, Contractor Registration & Empanelment, and Water E-Tariff & Billing — and with State e-Governance platforms (Jharkhand SSO, DigiLocker, etc.).
- Enable transparent publication of tenders, notifications, schemes, RTI disclosures, projects, achievements, photo/video galleries, news, and press releases.
- Provide structured citizen-services such as grievance redressal, RTI online filing, feedback, suggestion, and contact-us facilities.
- Ensure adherence to all applicable Government of India and Government of Jharkhand digital-governance standards, IT policies, and cyber-security mandates.

11.2 Scope of Work — Component-E

The selected Bidder shall be responsible for end-to-end design, development, implementation, hosting, integration, security audit, training, and maintenance of the CMS-based Departmental Website. The scope shall include:

- Detailed As-Is study of the existing WRD departmental website, content audit, information architecture review, and stakeholder consultation.

- Preparation and sign-off of Information Architecture (IA), Sitemap, Wireframes, UI/UX prototypes, and Content Migration Plan.
- Design and development of a fresh, modern, intuitive, and visually consistent website on an open-source CMS platform (e.g., Drupal 10+ or WordPress LTS as approved by the Department), GIGW 3.0-compliant, with full content-management capabilities.
- Migration of all existing relevant content (textual, multimedia, documents, archives) from the legacy website with structured taxonomy, metadata enrichment, and zero data loss.
- Bilingual (Hindi/English) implementation across all static and dynamic content, including a workflow for translation and bilingual publication.
- Integration with the four (4) other components of this RFP, with deep linking, SSO, and shared design language.
- Implementation of all citizen-service modules including Grievance Redressal, RTI, Feedback, Contact-Us, Subscription, and Newsletter.
- CERT-In empanelled VAPT before Go-Live and annually thereafter, including remediation of all observations.
- Hosting at the Jharkhand State Data Centre (JAP-IT) / NIC Data Centre, with HA/DR, daily/weekly backups, and 90-day retention.
- Training of departmental content authors, editors, publishers, and administrators on the CMS.
- Three (3) years of comprehensive Operation & Maintenance support post Go-Live as part of AMC.

11.3 Website Design & UI/UX Requirements

- Full compliance with GIGW 3.0 guidelines issued by NIC / MeitY, including layout, navigation, metadata, and content standards.
- Adherence to Government of Jharkhand and Government of India branding guidelines, including the State Emblem, the Government of Jharkhand logo, and the Departmental identity.
- Fully responsive, mobile-first design supporting Desktop, Laptop, Tablet, and Mobile devices, including small-screen feature phones where feasible.
- WCAG 2.1 Level AA accessibility compliance — screen-reader compatibility, keyboard navigation, high-contrast mode, adjustable font size, ARIA landmarks, and skip-to-content links.
- Full bilingual interface (Hindi & English) with one-click language toggle persisting across all pages, dynamic content, and forms; Unicode-compliant Hindi rendering.
- Intuitive navigation with mega-menus, breadcrumbs, contextual tooltips, sticky headers, and 'back to top' utilities.
- Modern visual design with optimised typography, accessible colour palette, and consistent iconography.
- Dark mode support, customisable themes, and Progressive Web App (PWA) capabilities for mobile users (offline reading of last-viewed content).
- Optimised performance — page load under 3 seconds on a 4G connection, Lighthouse score $\geq$ 85 across Performance, Accessibility, Best Practices, and SEO.

- Search Engine Optimisation (SEO) with proper meta tags, sitemap.xml, robots.txt, schema.org structured data, and Open Graph tags for social-media sharing.

11.4 Public-Facing Modules

11.4.1 Module A — Home Page & Landing Experience

- Dynamic, CMS-driven home page with configurable banners, sliders, and announcements.
- Quick-access tiles for key citizen services (Apply for Water Allocation, Pay Water Bill, Contractor Registration, Tenders, RTI, Grievance, etc.).
- Latest news ticker, scrolling notifications, and 'What's New' section with archive.
- Photo and video carousel showcasing departmental projects and achievements.
- Hon'ble Minister and Secretary message sections with dynamic update facility.
- Statistics widget showing key WRD metrics (number of dams, irrigation potential created, ongoing projects, etc.) with auto-update from PPMS where applicable.
- Configurable footer with sitemap, contact information, social-media links, and Government of Jharkhand quick-links.

11.4.2 Module B — About the Department

- Vision, Mission, Mandate, and History of the Department.
- Organisational structure with hierarchical chart (auto-rendered from CMS data).
- Profile pages of Hon'ble Minister, Secretary, Engineer-in-Chief, Chief Engineers, and key officers.
- Departmental functions, divisions, circles, and zones with downloadable contact directory.
- Citizens' Charter with downloadable PDF in Hindi and English.

11.4.3 Module C — Schemes, Projects & Programmes

- Catalogue of departmental schemes (Major, Medium, Minor Irrigation; Lift Irrigation; Watershed; Multi-Purpose Projects; etc.) with eligibility, benefits, application process, and downloadable forms.
- Project showcase with filters by district, river basin, scheme type, and implementation status.
- Live project status feed (read-only) sourced from PPMS via secure API.
- Photo and video galleries for each project.

11.4.4 Module D — Tenders & Notices

- Tender repository with filters by date, division, value, status (Open/Closed/Awarded), and tender type.
- Auto-deep-link integration with the State e-procurement portal (jharkhandtenders.gov.in).
- Corrigendum, Addendum, and Result-of-Tender publication with notification subscriber alerts (email/SMS).
- Archive of past tenders with full audit trail.
- Notifications, Office Orders, Circulars, and Government Resolutions section with category-wise filtering and search.

11.4.5 Module E — Acts, Rules & Policies

- Repository of Acts, Rules, Policies, Manuals, Codes, and Guidelines applicable to WRD.
- Year-wise, category-wise, and keyword-search filters.
- Bilingual versions of all documents wherever available.
- Version control and change-log for amended documents.

11.4.6 Module F — Citizen Services & Online Applications

This module shall serve as the unified citizen-services gateway, providing curated entry points and deep links to:

- Industrial Water Allocation Portal (Component-B) — apply, track, renew, pay.
- Contractor Registration & Empanelment Portal (Component-C) — register, renew, download certificate.
- Water E-Tariff & Billing Portal (Component-D) — view bill, pay bill, raise grievance.
- PPMS public dashboards (Component-A) — view project progress.
- Single Window Clearance System (SWCS) for inter-departmental services.

11.4.7 Module G — Grievance Redressal

- Online grievance submission with category, sub-category, location, and supporting document upload.
- OTP-based grievance authentication.
- Auto-routing of grievance to the concerned division/officer based on category and jurisdiction.
- SLA-based escalation matrix with auto-escalation on breach.
- Grievance tracking by reference number with SMS/email status updates.
- Public dashboard showing grievance disposal statistics (anonymised).
- Integration with State CPGRAMS / JharSeva, where applicable.

11.4.8 Module H — RTI Online

- Online filing of RTI applications and First Appeals as per the Right to Information Act, 2005.
- Online payment of RTI fees through JE-GRAS / e-payment gateway.
- Tracking of RTI applications by reference number.
- Suo-motu disclosure under Section 4(1)(b) with structured 17-point format and downloadable PDFs.
- Repository of RTI replies and decisions (with appropriate redaction of personal data).

11.4.9 Module I — Tenders, Recruitment & Career

- Recruitment notifications, advertisements, and result publication.
- Internship and engagement opportunities.
- Filter and search by post, division, qualification, and status.

11.4.10 Module J — Media & Communications

- Press releases with bilingual versions.

- Photo gallery with album, lightbox view, and date filters.
- Video gallery with embedded YouTube/Vimeo support and bandwidth-adaptive playback.
- Media coverage repository.
- Social-media integration (Twitter/X, Facebook, Instagram, YouTube, LinkedIn) with auto-fetch of latest posts.

11.4.11 Module K — Reports, Publications & Datasets

- Annual Reports, Statistical Hand-books, Master Plans, DPRs (where public), and Vision Documents.
- Open-data publication of non-sensitive datasets in Excel/CSV/JSON, in alignment with the National Data Sharing and Accessibility Policy (NDSAP) and data.gov.in.
- Subscription mechanism for newsletters and report releases.

11.4.12 Module L — Search, Sitemap & Help

- Powerful site-wide search with auto-suggest, typo tolerance, faceted filtering, and bilingual indexing.
- Auto-generated XML sitemap and HTML sitemap pages.
- Help, FAQ, and User-Guide sections with categorised search.
- Contact-Us page with division-wise contact directory and inline map.

11.4.13 Module M — Standard Mandatory Pages (GIGW)

- Privacy Policy, Terms of Use, Disclaimer, Copyright Policy, Hyperlinking Policy.
- Accessibility Statement and Help.
- Website Policies.
- Web Information Manager (WIM) details.
- Last Updated, Visitor Counter, and Page-Wise Last Reviewed metadata.

11.5 Content Management System (CMS) Requirements

The website shall be built on an enterprise-grade open-source CMS, with the following capabilities:

- Role-based access control with at least the following roles: Super Admin, Admin, Web Information Manager, Editor, Content Author, Translator, and Approver.
- WYSIWYG editor with support for rich text, images, embedded videos, tables, accordions, tabs, and reusable content blocks.
- Drag-and-drop page builder for non-technical users to create and update pages without coding.
- Workflow-based content publishing — Author → Reviewer → Approver → Publisher — with audit trail and rollback.
- Content scheduling (publish-on-date, expire-on-date) and version history with diff comparison.
- Bilingual content management with side-by-side English/Hindi editing and translation status tracking.
- Centralised media library with image optimisation, alt-text enforcement, and document versioning.

- Taxonomy management for categories, tags, and metadata.
- URL management with friendly slugs, redirects, and 301/410 mapping for migrated content.
- Form builder for creating dynamic public forms (feedback, subscriptions, surveys) with anti-spam and CAPTCHA.
- In-built SEO module for meta-data, canonical URLs, sitemap generation, and schema injection.
- Audit logs for every administrative and content-related action.
- CMS shall support upgrades, plug-ins, and themes within the AMC scope.

11.6 Integration Requirements (Component-E)

- Single Sign-On (SSO) with Jharkhand State SSO / SWCS, where applicable, so that registered users have a unified identity across the WRD digital ecosystem.
- API-based integration with PPMS (Component-A) for live project counts, project status, and dashboards (read-only public data).
- API-based integration with the Industrial Water Allocation Portal (Component-B) for application tracking and license verification (read-only).
- API-based integration with the Contractor Registration & Empanelment Portal (Component-C) for contractor verification and certificate validation (QR/ID lookup).
- API-based integration with the Water E-Tariff & Billing Portal (Component-D) for consumer bill view and payment redirection (read-only summary).
- Integration with the SMS Gateway (JAPIT/CDAC) for citizen alerts.
- Integration with the Government Email Service (DKIM/SPF compliant) for newsletters and notifications.
- Integration with DigiLocker for citizen certificate verification, where applicable.
- Embedding of GIS / Google Maps (or equivalent open-source map service) for project locations, dam locations, and division boundaries.
- Integration with the National Voluntary Blood Donation / Aspirational Districts Programme widgets where mandated for Government websites.
- Compliance with the National Portal of India (india.gov.in) hyperlinking requirements.

11.7 Bilingual & Accessibility Requirements

- Full Hindi (Devanagari) support across UI labels, dynamic content, dropdowns, dates, numbers, and search.
- Unicode 13+ rendering with mandatory inclusion of UNICODE-compliant Hindi fonts as per GIGW guidelines.
- Bilingual sitemap and bilingual search results.
- WCAG 2.1 Level AA compliance for both English and Hindi versions.
- Accessibility plug-ins / overlays for font size adjustment (A-/A/A+), high-contrast theme, and screen-reader hints.
- Cross-browser compatibility (Chrome, Edge, Firefox, Safari — last two major versions).
- Mobile responsiveness verified on Android and iOS for the latest two major versions.

11.8 Citizen Engagement & Analytics

- Visitor counter with month-wise, year-wise, and total counts displayed on the home page.
- Integration with web-analytics platforms — preferably MeitY-approved or open-source (e.g., Matomo).
- Heat-mapping and user-journey analytics for content optimisation.
- Newsletter subscription with double opt-in and unsubscribe support (CAN-SPAM/IT-Act compliant).
- Push-notification support via web (where opted in by the user).

11.9 Content Migration

The selected agency shall perform complete content migration from the existing departmental website to the new CMS-based website. The scope shall include:

- Content audit of all pages, documents, images, videos, and downloadable files on the legacy website.
- Categorisation, taxonomy mapping, and metadata enrichment for all migrated content.
- Identification of obsolete, duplicate, and broken content with the Department's approval before discard.
- Migration of historical archives (orders, notifications, circulars, tenders, photographs) with appropriate dating and indexing.
- Maintenance of legacy URL paths through 301 redirects to preserve external links and SEO equity.
- Re-OCR / re-tagging of scanned PDFs to make them searchable, where feasible.
- Reconciliation report at the end of migration certifying zero data loss and full integrity.

11.11 Security Requirements (Component-E)

- OWASP Top 10 vulnerability compliance.
- CERT-In empanelled VAPT before Go-Live and annually thereafter, with full closure of audit observations.
- Web Application Firewall (WAF) with bot mitigation.
- DDoS protection with automated mitigation.
- Captcha (image / reCAPTCHA / hCaptcha) on all public forms.
- Rate-limiting and IP-blacklisting for abusive traffic.
- Encrypted backups with 90-day retention and quarterly restoration drills.
- Hardened CMS configuration with disabled unused modules, secured admin URL, and 2FA for all CMS administrators.
- Compliance with the Information Technology Act, 2000 and the Digital Personal Data Protection Act, 2023.

11.12 Hosting, Backup & Disaster Recovery (Component-E)

- Hosting at the JAP-IT / NIC Data Centre as directed by the Department.
- High-Availability (HA) architecture with load balancing and failover.
- Disaster Recovery (DR) setup at a geographically separate location (where mandated).

11.13 Training (Component-E)

Training shall be conducted for at least ten (10) departmental personnel including content authors, editors, the Web Information Manager, and CMS administrators. Duration shall be a minimum of three (3) working days, supplemented by recorded video tutorials and a downloadable bilingual user manual. Topics shall include:

- CMS basics, content authoring, and the editorial workflow.
- Bilingual content publishing and translation workflow.
- Media library, taxonomy, and metadata management.
- Form builder and campaign management.
- User and role management.
- Backup, recovery, and basic security hygiene.

11.14 Deliverables (Component-E)

- Information Architecture, Sitemap, and Wireframes.
- UI/UX clickable prototypes (high-fidelity) with bilingual screens.
- Software Requirement Specification (SRS) document.
- System Design Document (SDD), including HLD and LLD.
- Content Migration Plan and Reconciliation Report.
- Source code, theme, and custom-module documentation.
- CMS Administrator Manual (bilingual).
- Editorial / Content-Author Manual (bilingual).
- Test Plan, Test Cases, and Test Results.
- CERT-In VAPT Compliance Certificate.
- GIGW 3.0 Compliance Certificate / NIC GIGW Audit Report.
- WCAG 2.1 Level AA Accessibility Audit Report.
- Go-Live Report and Post-Implementation Review.
- Quarterly performance, uptime, and security reports during AMC.

11.15 Implementation Timeline (Component-E)

Activity	Timeline
As-Is study, IA, Sitemap, Wireframes & Prototype Sign-off	30 Days
CMS Setup, Theme Development, Module Development	60 Days

Activity	Timeline
Content Migration, Bilingual Content Creation	30 Days
Integration with Components A, B, C, D and external systems	20 Days
Testing (SIT, UAT) & GIGW / WCAG / VAPT Compliance	20 Days
Training, Soft Launch & Go-Live	10 Days

Total project duration for Component-E: Approximately 5–6 months from sign-off of SRS, to be aligned within the overall 6-month integrated development window.

11.16 Ownership (Component-E)

- All content, source code, custom themes, custom modules, and database shall remain the sole property of WRD.
- Visitor analytics data, user accounts, grievance data, and any personal data collected through the website shall remain the sole property of WRD.
- All licences for proprietary CMS plug-ins (if any) procured under this scope shall be in the name of WRD.

11.17 Expected Benefits (Component-E)

- Unified, modern, and trustworthy digital identity for the Water Resources Department.
- Higher citizen engagement through accessible and bilingual content.
- Self-service content publishing by the Department, reducing dependence on IT vendors.
- Faster and transparent dissemination of tenders, notifications, and schemes.
- Improved compliance with GIGW 3.0, WCAG 2.1, and CERT-In standards.
- Strengthened citizen trust through grievance redressal, RTI, and feedback mechanisms.
- Search-engine visibility and improved discoverability of departmental information.

12. Data Migration

The selected Agency shall be responsible for complete, accurate and verifiable data migration from the existing legacy systems (including but not limited to wrdd.jharkhand.gov.in, wrdreg.jharkhand.gov.in, the existing PPMS database, the existing E-Tariff/Billing records and the existing departmental website) to the newly developed integrated solution. Data migration shall be a deliverable in its own right and shall be signed off by the Department before final Go-Live of each component.

12.1 Scope of Data Migration

- Migration of all applicant/contractor registration data including personal details, addresses, contact information, PAN/GST/Aadhaar references, organizational profile and complete registration history.
- Migration of all uploaded documents (photographs, signatures, PAN cards, address proofs, affidavits, incorporation certificates, partnership deeds, MOA/AOA, sanction orders, agreements, allotment letters, etc.) with proper logical mapping and integrity verification using checksums.
- Migration of all application records covering acknowledgement numbers, application dates, contractor classes, current status, complete file-movement history and stage-wise remarks/timestamps with officer identifiers.
- Migration of all certificate records — certificate numbers, issuance dates, validity periods, QR codes, digital signatures, revocation history and any associated supporting documents.
- Migration of all financial records — payment transactions, challan numbers, GRN numbers, JE-GRAS references, head-of-account, scheme codes, refund records and reconciliation history.
- Migration of water allocation records — source, sanctioned quantity, validity, renewal history, billing records, meter readings, drawal entries, tariff slabs, division mapping and inspection data.
- Migration of E-Tariff billing data — consumer master, agreement details, division-wise mapping, demand notices, payment history, arrears, penalty/interest ledgers and recovery/defaulters status.
- Migration of all departmental website content — pages, menus, news/notices, tenders, schemes, photo and video galleries, downloadable documents — with bilingual content preserved and metadata retained.
- Migration of all blacklist records, master data, user accounts and credentials of departmental officers (with mandatory forced password reset on first login of the new system).

12.2 Migration Methodology

- Data Audit: A full audit of existing data sources, schemas, file stores and content shall be conducted in the first 30 days of the contract and an audit report shall be submitted to the Department.
- Data Cleansing & Deduplication: Inconsistencies, duplicate records, orphan documents and invalid entries shall be identified and addressed in consultation with the Department.
- Mapping Document: A field-by-field source-to-target mapping document shall be prepared, version-controlled and signed off by the Department before any production migration.

- **Mock Migration:** A minimum of two full mock migration cycles shall be executed in a staging environment with reconciliation reports shared with the Department.
- **Reconciliation Reports:** Record counts, financial totals, stage-wise pendency totals and document-count tallies shall be reconciled between source and target. Zero data loss is the mandatory acceptance criterion.
- **Parallel Run:** A minimum 30-day parallel run of the legacy and new systems shall be carried out for each component before switch-over, during which any deltas shall be migrated daily.
- **Roll-back Plan:** A complete documented roll-back plan shall be in place for each cut-over.
- **Sign-off:** Final cut-over shall be performed only after written approval from the Department on the Reconciliation Sign-off Certificate.

13. Technology Stack & Technical Requirements

The Agency shall use a modern, scalable, secure and well-supported technology stack across all five components. The minimum technical requirements set out below shall be adhered to. The Agency may propose superior alternatives, subject to written approval of the Department.

Layer / Component	Minimum Specification
Frontend	React 18+ (LTS) with TypeScript, Tailwind CSS 3+ / Bootstrap 5, mobile-first responsive design, Progressive Web App (PWA) ready.
Backend (PPMS — Component A)	Node.js 20 LTS (Express / NestJS) OR Django 5 LTS (Python) OR Laravel 11 (PHP 8.3) — open-source frameworks with no proprietary licensing cost, with PostgreSQL; OR .NET / .NET Core (MVC or API-first) as a permitted alternative subject to written approval of the Department.
Backend (Components B, C, D)	Node.js 20 LTS (Express / NestJS) OR Django 5 LTS (Python) OR Laravel 11 (PHP 8.3) — open-source frameworks with no proprietary licensing cost, with PostgreSQL; OR .NET / .NET Core (MVC or API-first) as a permitted alternative subject to written approval of the Department.
Backend (CMS — Component E)	WordPress 6.x LTS on PHP 8.2+ with hardened configuration, role-based authoring and bilingual taxonomy; OR Strapi (headless CMS on Node.js); OR Laravel 11 (PHP 8.3).
Database	PostgreSQL 16 (primary), with MariaDB 11 as a permitted alternative — including normalization, indexing, query optimization, stored procedures, transaction safety and audit tables.
Application Server	Nginx (reverse proxy with SSL termination) with PM2 (Node.js) / Gunicorn (Python) / PHP-FPM as application runtimes; Apache as a permitted alternative.
Operating System	Ubuntu 22.04 LTS or 24.04 LTS for all server workloads.
Caching & Search	Redis 7 for caching; OpenSearch or Meilisearch for full-text search and analytics.

Layer / Component	Minimum Specification
Security	TLS 1.3, WAF, OWASP Top 10 compliance, CERT-In empanelled VAPT, AES-256 encryption at rest, MFA, RBAC, secure key vault for secrets.
Hosting	NIC / Jharkhand State Data Centre (JH-SDC) / JAPIT / MeitY-empanelled cloud as directed by the Department, with HA and DR.
SMS Gateway	JAPIT SMS Gateway / CDAC / Government-approved SMS service with DLT compliance.
Email Service	JAPIT email / Government-approved email with DKIM, SPF and DMARC.
Digital Signature	CCA-approved DSC provider integration; Aadhaar e-Sign integration where permitted.
CDN & Optimisation	Lazy loading, asynchronous API rendering, GZIP/Brotli compression and CDN-ready static assets, including Cloudflare or BunnyCDN (or any equivalent Government-approved CDN).
Version Control	Git (self-hosted Gitea / GitLab CE, or Government-approved equivalent) with a defined branching strategy and CI/CD pipelines (GitHub Actions / GitLab CI or equivalent).
Mobile Application	React Native 0.74+ — single codebase for Android and iOS, reusing the React frontend skill base.

14. Security Requirements & Compliance

The Agency shall implement comprehensive security across all five components in line with prevailing Government of India and Government of Jharkhand security standards. Security is non-negotiable and any breach of these requirements shall constitute a material breach of contract.

14.1 Application Security

- Compliance with the OWASP Top 10 across all web and mobile applications, evidenced by static and dynamic analysis reports.
- Vulnerability Assessment and Penetration Testing (VAPT) by a CERT-In empanelled auditor before initial Go-Live and annually thereafter for the full contract duration.
- Multi-Factor Authentication (MFA) for all departmental users; OTP-based authentication for citizens and contractors.
- Role-Based Access Control (RBAC) implemented under the principle of least privilege, with periodic access review.
- API authentication using OAuth 2.0 / signed JWT, rate-limiting, input validation and strict CORS policies.

- Password storage using bcrypt or argon2id with strong work factors; mandatory password policy and lockout on repeated failed attempts.
- Comprehensive audit logging of all create, update, delete and approval actions with tamper-evident storage.

14.2 Infrastructure Security

- Web Application Firewall (WAF) deployed in front of every public-facing application.
- DDoS protection with automated mitigation and traffic-anomaly alerting.
- Critical security patches applied within 48 hours of vulnerability disclosure; non-critical patches under a defined cadence.
- 24x7 monitoring of all production assets with alert escalation for security incidents and SOC integration where available.

14.3 Regulatory & Policy Compliance

- Information Technology Act, 2000 and all amendments thereof, including the IT (Reasonable Security Practices and Procedures) Rules.
- Digital Personal Data Protection Act, 2023 (DPDP Act) in respect of all personal data handled by the system.
- All applicable CERT-In advisories, directions and reporting timelines, including the 6-hour incident reporting requirement.
- Aadhaar Act and Regulations and UIDAI guidelines for any e-KYC integrations.
- Guidelines for Indian Government Websites (GIGW 3.0), MeitY guidelines on cloud and the Government of Jharkhand IT Policy.
- Government of India National Cyber Security Policy, NIST Cybersecurity Framework alignment and the latest applicable Standards (BIS) where relevant.

15. Hosting, Deployment & Infrastructure

The Agency shall be responsible for hosting and deployment of all five components on infrastructure compliant with Government of India guidelines, as directed by the Department.

- All applications shall be hosted at NIC Data Centre / Jharkhand State Data Centre (JAP-IT) / cloud, as directed by the Department, on adequately sized web, application, database, backup and staging environments.
- High Availability (HA) architecture with redundant nodes, load balancing and automated failover.
- Disaster Recovery (DR) site in a geographically separate location, with Recovery Point Objective (RPO) of 4 hours or better and Recovery Time Objective (RTO) of 8 hours or better; DR drill at least once every six months.
- Automated backup system — daily incremental and weekly full — with minimum 90-day retention and quarterly restore-tests.
- EV-class SSL certificates from a Government-approved Certificate Authority, with automated renewal monitoring.

- DNS configuration and domain management for all departmental domains and sub-domains used by the integrated solution.
- CDN deployment for static content optimization and protection of public-facing endpoints.
- Mirror staging and UAT environments matching production, used for testing every change before promotion.
- Centralized monitoring with 24x7 automated alerts for server health, application errors, database performance and security incidents, with on-call escalation matrix.
- Mandatory CERT-In security audit clearance before initial hosting, and after every major release at JAP-IT, with patching of all audit observations within agreed timelines.

16. Timelines & Deliverables

Macro-level timelines for the integrated engagement are as set out below. Component-specific timelines are detailed within each component's scope.

16.1 Macro Timeline

Phase	Indicative Duration
On-site Manpower Deployment as per the entire scope of work for the design, development and implementation of all components (A, B, C, D and E).	T0 + 2
Project Inception, Kick-off and Mobilization	T0 + 1 weeks
As-Is Study, Requirement Gathering and SRS for all five components	T0 + 8 weeks
System Design Document (HLD/LLD) and UI/UX Prototypes	T0 + 16 weeks
Phase-1 Development: PPMS, Industrial Water Allocation, Contractor Registration	T0 + 36 weeks (5 months)
Phase-1 UAT: PPMS, Industrial Water Allocation, Contractor Registration	T0 + 40 weeks (6 months)
Phase-2 Development: Water E-Tariff & Billing Portal and CMS Departmental Website	T0 + 46 weeks (8 months)
Phase-2 UAT: Water E-Tariff & Billing Portal and CMS Departmental Website	T0 + 48 weeks (9 months)
Final Integrated Go-Live, VAPT clearance, Training completion and HOTO	T0 + 52 weeks (12 months)
Operation & Maintenance (AMC)	3 (three) years from Final Go-Live
Total Contract Duration	4 years (extendable up to 2 years)

16.2 Master Deliverables

S.No.	Deliverable	Phase
1	Project Inception Report with detailed Work Plan and Communication Matrix	Post-Agreement
2	As-Is Study Report covering all five legacy systems	Requirements Phase
3	Software Requirement Specification (SRS) — component-wise and consolidated	Requirements Phase
4	System Design Document (SDD) including HLD and LLD	Design Phase
5	UI/UX Design Mockups and Clickable Prototypes (Hindi & English)	Design Phase
6	Data Migration Plan, Mapping Document and Mock-Migration Reports	Pre-Deployment
7	Complete Source Code with technical documentation and API specifications	Development Phase
8	Test Plan, Test Cases and Test Results (SIT, UAT, Performance, Load)	Testing Phase
9	VAPT Report with CERT-In Compliance Certificate (initial and annual)	Pre Go-Live & AMC
10	Data Migration Execution Report and Reconciliation Sign-off	Pre-Deployment
11	UAT Sign-off Report from the Department	Pre Go-Live
12	User Manuals (Hindi & English) — Admin, Citizen, Officer, System Admin	Training Phase
13	Training Materials, Presentations, Tutorials and Train-the-Trainer Report	Training Phase
14	Go-Live Report and Post-Implementation Review	Deployment
15	Complete Source Code Handover with all IPR documentation and HOTO certificate	Post Go-Live
16	Quarterly AMC Health Reports, MIS Reports and SLA Compliance Reports	AMC Period

17. Project Management Unit (PMU), Operation & Maintenance

For the Operation & Maintenance (O&M) of the integrated software solution covering all five components, the Agency shall establish a dedicated Project Management Unit (PMU) with on-site technical manpower at the Department's premises in Ranchi. The PMU shall ensure smooth functioning, ongoing development, change-request handling, security patching and SLA-based support throughout the contract.

17.1.1 Role and responsibility of Deployed PMU Manpower.

17.1 Activities During AMC

- New feature roll-outs and version upgrades on agreed change-request lines.
- Compatibility upgrades for operating system, browsers, application server and database.
- Timely issue resolution under SLAs, backup support, environment refresh and re-installation as needed.
- Annual VAPT and patching of audit observations within agreed timelines.
- All change requests (CRs) to follow formal documentation and approval through a CR register.
- Quarterly health reports, monthly MIS, weekly status calls and a daily on-site presence.

17.2 Mandatory On-Site Key Personnel

The PMU shall include, as a minimum, the following on-site key personnel deployed at WRD Ranchi for the full AMC duration. Educational qualifications, experience and roles are summarised below; full CVs shall be furnished as per Form T-4.

Position	Min. Qualification	Min. Experience	Indicative Strength
Project Manager (PM)	MCA / B.E. / B.Tech / MBA	10 years	1
Sr. Software Developer	MCA / B.E. / B.Tech	5 years	1
Software Developer	MCA / B.E. / B.Tech	2 years	1
Jr. Software Developer	BCA / MCA / B.E. / B.Tech	1 year	1
Mobile App Developer	MCA / B.E. / B.Tech	2 years	1
UI/UX Designer	B.Des / MCA / B.E. / B.Tech	3 years	1
Quality Assurance / Test Engineer	MCA / B.E. / B.Tech	3 years	1

17.3 Roles & Responsibilities — Key Roles

(i) Project Manager

- Overall accountability for delivery, governance and stakeholder management across all five components.
- Ownership of the project plan, risk register, change-request log and reporting cadence with the Department.
- Single point of contact for the Department for all contractual and delivery matters.

(ii) Sr. Software Developer

- Lead the software development team and act as primary technical point of contact.
- Translate requirements into design and oversee implementation, code-reviews and best-practice enforcement.
- Ensure smooth functioning of all five components, address issues promptly and oversee custom-report generation.
- Coordinate with other team members on technical challenges and provide expertise on system upgrades.

(iii) Software Developer

- Develop new features and modules within the software, including coding, debugging and unit testing.
- Generate financial and operational reports as required by the Department.
- Assist the Sr. Developer in implementing system upgrades and enhancements.
- Ensure stability and performance of all software components.

(iv) Jr. Software Developer

- Assist the development team in coding, testing and debugging.
- Work under guidance of Sr. and Software Developers to implement features.
- Participate in troubleshooting and resolution and contribute to minor system updates and reports.

(v) Mobile App Developer

- Develop and maintain the mobile applications associated with the integrated solution (Android & iOS, ideally cross-platform such as Flutter or React Native).
- Customise and update the mobile app as per project requirements; ensure integration with backend APIs.
- Coordinate with the Sr. Developer to ensure feature parity and consistent UX across web and mobile.

17.4 SLA Framework During AMC

Detailed SLA-based penalties applicable during AMC are set out in Section II — Special Conditions. The Agency shall publish a monthly SLA compliance dashboard to the Department.

Annexure-1: Format for Financial Bid

To,
The Executive Engineer,
Water Ways Division, Ranchi,
Water Resources Department,
Government of Jharkhand.

Subject: Selection of Agency for Re-Design, Re-Development, Implementation, Operation and Hosting of an Integrated Software Solution for the Water Resources Department, Government of Jharkhand, comprising Project Progress Monitoring Portal (PPMS), Contractor Registration & Empanelment Portal, Industrial Water Allocation Portal, Water E-Tariff & Billing Portal, and CMS-based Departmental Website, along with Comprehensive Maintenance and Support for a period of three (3) years.

Reference: This office NIT No.: WRD/JH/CE/DMPH/RFP/_____/2025-26

I/We hereby submit our Financial Bid for the captioned assignment in accordance with the norms, specifications and conditions stipulated in the Scope of Services and at the rates quoted online in the Financial Bid section of the Tender. The component-wise indicative break-up of our quoted price is furnished below; the total of Section-A and Section-B shall be deemed to be the overall bid price for evaluation under QCBS. All amounts are exclusive of applicable GST, which shall be paid extra as per prevailing rates.

SECTION A — One-Time Development & Implementation

S.No.	Component / Deliverable	Amount (INR, excl. GST)
A1	Component-A: Project Progress Monitoring Portal (PPMS) — Take-over of existing portal, development of Fund Requisition Module, enhancements to Reporting/MIS/Dashboards, Google Maps & SMS Gateway integration, security audit, hosting & deployment	
A2	Component-B: Industrial Water Allocation Portal (wrdd.jharkhand.gov.in) — Re-engineering & modernisation, SWCS & JE-GRAS integration, multi-level approval workflow, billing & revenue, dashboards & MIS, security audit, hosting & deployment	
A3	Component-C: Contractor Registration & Empanelment Portal (wrddreg.jharkhand.gov.in) — Complete redesign and redevelopment, public modules, department admin panel, certificate management with QR & DigiLocker, payment integration, security audit, hosting & deployment	

S.No.	Component / Deliverable	Amount (INR, excl. GST)
A4	Component-D: Water E-Tariff & Billing Portal — Consumer registration, water-drawal entry, role-based JE→AE→EE workflow, tariff & billing engine, JE-GRAS / treasury / bank gateway integration with division-wise revenue credit, dashboards & MIS, security audit, hosting & deployment	
A5	Component-E: CMS-based Departmental Website — GIGW 3.0/WCAG 2.1 AA bilingual website, CMS configuration, public modules, content migration, citizen-services gateway, security audit, hosting & deployment	
A6	Cross-cutting: Data migration, integration testing, UAT, VAPT (initial), CERT-In compliance, training, documentation, HOTO and Final Integrated Go-Live	
	Sub-total (A) — One-Time Development & Implementation	

SECTION B — Operation, Maintenance & Support (AMC) for 3 years

S.No.	AMC Component (per annum, all components combined)	Amount (INR, excl. GST)
B1	AMC Year-1: PMU on-site manpower, corrective & preventive maintenance, change requests, OS/DB/browser compatibility upgrades, monitoring, support and SLA compliance — across all five components	
B2	AMC Year-2: As above (with up to mutually agreed escalation, if any)	
B3	AMC Year-3: As above (with up to mutually agreed escalation, if any)	
B4	Annual VAPT for 3 years (one cycle per year by CERT-In empanelled auditor)	
	Sub-total (B) — AMC for 3 years	

GRAND TOTAL

	Description	Amount (INR, excl. GST)
	Sub-total (A): One-Time Development & Implementation	
	Sub-total (B): AMC for 3 years	
	GRAND TOTAL (A + B) — Total Quoted Bid Price for QCBS evaluation	

Total Quoted Bid Price in figures: INR _____

Total Quoted Bid Price in words: Rupees _____ only.

I/We hereby promise to submit a detailed cost break-up and analysis for the consultancy work, when asked by the Department. I/We further confirm that:

29. The above prices are firm and shall remain valid for a period of 120 days from the last date of bid submission.
30. The prices are inclusive of all costs of personnel, infrastructure, licences (other than third-party paid licences explicitly excluded), travel, lodging, communications and any other expenses required to deliver the scope of work.
31. Applicable GST and any other statutory taxes shall be paid extra as per prevailing rates and shall be invoiced separately.
32. In case of any discrepancy between the figures and words, the amount in words shall prevail.

Signature of Witness

Name: _____

Designation: _____

Signature of Authorised Signatory

Name: _____

Designation: _____

For and on behalf of M/s _____

Telephone No.: _____ Fax No.: _____ Email: _____

Date: _____

Place: _____

(Affix Seal of the Agency/Bidder/Consultant)

Annexure-2: Performance Bank Guarantee

(To be issued by a Scheduled Commercial Bank on non-judicial stamp paper of appropriate value, on the Bank's letterhead)

To,
The Executive Engineer,
Water Ways Division, Ranchi,
Water Resources Department,
Government of Jharkhand.

Whereas the Water Resources Department, Government of Jharkhand (hereinafter referred to as the "Department") has invited bids vide NIT No. WRD/JH/CE/DMPH/RFP/_____/2025-26 for Selection of Agency for Re-Design, Re-Development, Implementation, Operation and Hosting of an Integrated Software Solution comprising Project Progress Monitoring Portal (PPMS), Contractor Registration & Empanelment Portal, Industrial Water Allocation Portal, Water E-Tariff & Billing Portal, and CMS-based Departmental Website, along with Comprehensive Maintenance and Support for a period of three (3) years (hereinafter referred to as the "Project").

And whereas M/s _____ (hereinafter referred to as the "Service Provider") having its registered office at _____ has submitted its proposal in response to the said bid; and the Department has decided to award the contract to the Service Provider vide its Letter of Intent No. _____ dated _____ directing the Service Provider to submit a Performance Bank Guarantee for an amount of Rs. _____ (Rupees _____ only).

The Service Provider has requested us, _____ Bank (hereinafter referred to as the "Bank"), to furnish the above Bank Guarantee, valid up to _____ in favour of the Department.

Now, considering the request, we, _____ Bank, do hereby unconditionally and irrevocably undertake to pay to the Water Resources Department, Government of Jharkhand, an amount not exceeding Rs. _____ (Rupees _____ only) on demand by the Department, in case of failure of M/s _____ in fulfilling its obligations properly and timely under the said contract.

We, _____ Bank, do hereby further undertake to pay the amount payable under this Guarantee without demur, merely on a demand from the Department. Any such

demand made on the Bank shall be conclusive as regards the amount due and payable by the Bank under this Guarantee.

Our liability under this Guarantee shall be restricted to an amount not exceeding Rs. _____ (Rupees _____ only). This Guarantee shall remain in full force up to _____ unless the demand or claim under this Guarantee is made on us in writing on or before _____, after which we shall be discharged from all liabilities under this Guarantee.

We undertake to pay to the Department any money so demanded notwithstanding any dispute or disputes raised by the Service Provider in any suit or proceeding pending before any court or tribunal relating thereto, our liability under this Guarantee being absolute and unequivocal. The payment so made by us under this Guarantee shall be a valid discharge of our liability for payment thereunder, and the Service Provider shall have no claim against us for making such payment.

This Guarantee will not be discharged due to any change in the constitution of the Bank or the Service Provider.

We undertake not to revoke this Guarantee during its currency except with the prior consent of the Department in writing.

Notwithstanding anything herein contained, our liability under this Guarantee shall:

33. Be limited to a sum of Rs. _____ (Rupees _____ only).

34. Stand completely discharged and all our rights under this Guarantee shall stand extinguished if no claim or demand is made upon us in writing on or before _____.

Date: _____

Place: _____

Signature for and on behalf of the Bank

Name: _____

Designation: _____

Bank Seal:

Form T-1: Agency/Bidder/Consultant Details

S.No.	Particulars	Details
1	Name of Agency / Bidder / Consultant	
2 (a)	Address of Agency / Bidder / Consultant	
2 (b)	Phone Number	
2 (c)	Fax Number	
2 (d)	E-mail	
2 (e)	Website	
3 (a)	Name of Authorised Signatory to bid	
3 (b)	Designation	
3 (c)	Phone (Landline) / Phone (Mobile)	
3 (d)	Fax	
3 (e)	E-mail	
4	Name, address, telephone, fax and email at which communication is to be sent in respect of the bid	
5	Names of present Proprietors / Partners / Board of Directors	
6	Year of Incorporation / Establishment	
7	PAN	
8	GSTIN (with Jharkhand local registered address, where applicable)	
9	ISO Certifications held (9001:2015 / 27001:2018 / 20000-1)	
10	CMMI Level 3 status (with Appraisal Reference) — if held	
11	GMDIC Verified MSME Certification (Government of Jharkhand)	
12	Empanelment with State / Central Government / PSU (with letter reference)	

Signature of Authorised Signatory: _____

Name & Designation: _____

Date: _____

Place: _____

(Seal of the Firm / Company)

Form T-2: Past Experience

The firm's experience over the last three (3) financial years (2022-23, 2023-24, 2024-25) in the field of assignments of similar nature. (Please enclose supporting documents — work orders and completion certificates issued by the competent authority of the concerned Government Sector / PSU, not below the rank of Executive Engineer.)

S.No.	Name of Project	Nature of Project	Name of Department / Client	Total Cost of Project (INR)	Date of Commencement	Status (Completed / Ongoing)	Remarks
1							
2							
3							
4							
5							

(Signature of Agency/Bidder/Consultant and Seal)

Form T-3: Personnel Qualification & Competence

Qualification and competence of the personnel proposed for the assignment. CVs of all proposed Key Personnel shall be enclosed in the format prescribed in Form T-4. No Key Personnel shall have attained the age of 62 (sixty-two) years on the date of submission of the proposal.

S.No.	Name of Person	Proposed Position	Academic Qualification	Total Experience (Years)	Relevant Experience (Years)	Remarks
1		Project Manager				
2		Sr. Software Developer-1				
3		Mobile App Developer-1				
4		Software Developer-1				
5		Jr. Software Developer-2				
6		UI/UX Designer				

(Signature of Agency/Bidder/Consultant and Seal)

Form T-4: Format of Curriculum Vitae (CV) for Proposed Key Staff

1. Proposed Position: _____

2. Name of Staff: _____

3. Date of Birth: _____

4. Nationality: _____

5. Educational Qualifications

(List all academic qualifications, university/institution, year of award and division/grade.)

6. Employment Record

List all positions held by the staff member since graduation, giving dates, names of employing organisations, titles of positions held and locations of assignments. For each specific assignment, the experience period must be clearly mentioned along with a supporting certificate (specifically for the Project Manager / Team Lead).

7. Summary of the CV

Furnish a precise and accurate summary as below. The information in the summary will have a direct bearing on the evaluation of the CV.

Education

Particulars	Details
(i) Field of Graduation and Year	
(ii) Field of Post-graduation and Year	
(iii) Any other specific qualification	

Experience

Particulars	Years
(i) Total experience in software development and IT sector	
(ii) Responsibilities held	
(iii) Relevant experience for this assignment	

Certification

35. I am willing to work on the project, and I will be available for the entire duration of the project assignment, and I will not engage myself in any other assignment during the duration of this assignment on the project.

36. I, the undersigned, certify that to the best of my knowledge and belief, this CV correctly describes me, my qualifications, and my experience.

Signature of the Candidate: _____

Place: _____

Date: _____

Signature of the Authorised Representative of the Firm: _____

Name & Designation: _____

Place: _____

Date: _____

(Seal of the Firm / Company)

Form T-5: Affidavit

(To be executed on Indian Non-Judicial Stamp Paper of Rs. 100/- and duly notarised)

I, _____ S/o / D/o _____
Director / Authorised Signatory of M/s _____ having its registered
office at _____, do hereby solemnly affirm and declare as follows:

37. That I have been authorised to execute this affidavit on behalf of the company by the Board of Directors vide its resolution passed on _____.
38. That the Water Resources Department, Government of Jharkhand, vide advertisement published in _____, had invited bids from agencies/consultants for Selection of Agency for Re-Design, Re-Development, Implementation, Operation and Hosting of an Integrated Software Solution comprising Project Progress Monitoring Portal (PPMS), Contractor Registration & Empanelment Portal, Industrial Water Allocation Portal, Water E-Tariff & Billing Portal, and CMS-based Departmental Website, along with Comprehensive Maintenance and Support for a period of three (3) years.
39. That in response to the said advertisement, our firm has submitted its proposal to the Water Resources Department, Government of Jharkhand.
40. That the proposal of our firm M/s _____ contains true and correct information and particulars furnished as per the tender document.
41. That our firm has neither failed to perform on any contract, as evidenced by imposition of a penalty by an arbitral or judicial authority or a judicial pronouncement or arbitration award against our firm, nor has our firm been expelled from any project or contract by any public authority, nor has had any contract terminated by any public authority for breach on our part.
42. That our firm is not currently under Corporate Debt Restructuring (CDR) or Strategic Debt Restructuring (SDR) under the Bureau of Industrial and Financial Reconstruction (BIFR) or any successor authority.
43. That our firm is not blacklisted or debarred by any Central / State Government, Public Sector Undertaking, autonomous body or local authority in India as on the date of submission of the bid.

That the statements made in paragraphs 1 to 7 of the foregoing affidavit are true to my knowledge and belief, and if anything is found contrary, I stand liable to be prosecuted under the appropriate Acts and laws in force.

Solemnly affirmed by the said _____ at
_____ on this the _____ day of _____ 2025/2026.

Deponent

Identified by:

Name: _____

Address: _____

Notarised by:

(Signature, Seal and Registration Number of Notary Public)

--- END OF DOCUMENT ---